

Downstream Foundations of Upstream Collusion*

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Abstract

Retailers' expectations can determine whether a cartel among their suppliers takes hold. We study manufacturer collusion under secret two-part tariffs with competing retailers. Retailer beliefs can create a barrier to collusion distinct from manufacturers' incentive constraints: beliefs that sustain an established cartel may prevent its formation. Belief adjustment in response to observed conduct can overcome this barrier, but faster adjustment also makes coordinated deviations more attractive. Under suitable conditions, the cost of rebuilding retailer confidence after deviations can sustain collusion without an explicit punishment phase. Our results distinguish cartel formation from cartel persistence.

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1 Introduction

Collusion in vertically related markets is usually studied as an incentive problem among upstream firms: can manufacturers resist the temptation to deviate from a collusive agreement? This perspective overlooks the following problem. Manufacturers who secretly start an illegal cartel must increase their wholesale prices for retailers that are often initially unaware of the coordination. These retailers may, however, refuse to accept higher prices for fear of being outcompeted by fellow retailers that (continue to) receive better offers. When supply contracts are bilateral and secret, a retailer cannot verify the terms granted to its rival. An upstream cartel may consequently be sustainable once established, yet difficult or impossible to initiate.

The German coffee cartel illustrates this problem. In an initial phase, coffee roasters coordinated wholesale-price increases, but the arrangement was short-lived and resulted in only limited overcharges. The cartel became more stable and more effective after the coffee roasters combined these increases with resale price maintenance and coordination with retailers (Holler and Rickert, 2022). Related implementation problems arise in hub-and-spoke cases and in other supply-chain cartels in which vertical communication or pricing restraints accompany upstream coordination (Harrington, 2018; Garrod et al., 2021; OECD, 2019; Bundeskartellamt, 2016). Such practices may do more than communicate prices or facilitate monitoring. They may reassure retailers that higher input prices apply to all retailers rather than to one alone.

These cases reveal three stylized facts that motivate our analysis. Upstream coordination often has to be implemented through downstream firms: manufacturers affect retail prices only through the contracts retailers accept and the prices they set in response. Vertical communication and vertical restraints can create the mutual understanding needed for this implementation by reassuring retailers that comparable terms apply to their rivals. Moreover, cartel formation and cartel stability are distinct: an arrangement may be sustainable once established even though the transition from competition to collusion is difficult or fails altogether. Our model offers a common framework for these observations.

The third fact points to a distinction that is largely absent from the theory of collusion. The standard repeated-game incentive constraint asks whether collusion is sustainable once the colluding firms understand the collusive strategy. In a vertically related market, there is a separate question: Does there exist a path from a non-collusive history to a collusive outcome along which downstream firms are willing to accept and implement the relevant changes, such as the higher input and sales prices? This condition is distinct from the conventional stability condition. Sustainability concerns the incentives of manufacturers after collusion has been reached, whereas formability concerns whether the market can reach that outcome in the first place.

We study how retailers' beliefs determine both the formation and the stability of upstream collusion under secret contracting. The logic is as follows. The success of a cartel depends not only on the contracts offered by the manufacturers, but also on what each retailer expects

the rival retailer to have been offered. If retailers do not expect an industry-wide increase in wholesale prices, their contract acceptance and pricing decisions may prevent manufacturers from implementing one. Conversely, retailers that fully understand the collusive strategy may make an established cartel sustainable, but their beliefs need not support a transition to collusion after a history of competition.

Formally, we analyze an infinitely repeated game with two exclusive manufacturer--retailer pairs. The manufacturers are long-lived and privately offer two-part tariffs to their respective retailers. Each retailer observes its own contract offer but not the contract offered to the competing retailer. The retailers then decide whether to accept and compete in retail prices. At the end of each period, contracts and market actions become publicly observable, so past conduct can affect subsequent beliefs. Retailers maximize current profits and therefore do not themselves collude, which allows us to isolate their role in implementing upstream coordination.

Within a period, we maintain the usual passive-belief assumption: an unexpected offer by one manufacturer does not reveal the contemporaneous offer made by the other manufacturer (Segal, 1999; Rey and Vergé, 2004). Across periods, however, retailers may revise their assessment of whether manufacturers' wholesale prices are competitive or collusive in response to observed conduct. We use a parametric framework of beliefs that covers several distinct cases. Retailers may hold beliefs that do not react to past conduct, may fully anticipate the manufacturers' collusive strategies, or may gradually adapt, interpreting repeated high wholesale-price offers as evidence of an industry-wide collusive regime.

Our first result is negative. Suppose retailer beliefs do not react to the history of manufacturer conduct. With efficient two-part tariffs, manufacturers cannot sustain wholesale prices above the competitive level, regardless of how patient they are or which collusive strategies they use. The beliefs associated with the competitive outcome are self-fulfilling: a retailer expects its rival to face competitive terms, prices accordingly, and a manufacturer that raises its own wholesale price only loses sales. Because beliefs do not respond to past conduct, retail prices are the same whether manufacturers collude or revert to competition, so no future change in manufacturer conduct can punish a deviation.

This result differs from the familiar commitment problem of an upstream monopolist, such as in Hart and Tirole (1990). There, a manufacturer cannot commit to its own future offers. Here, in addition, each manufacturer is also unable to commit on behalf of the other manufacturer to offering comparable terms to the rival retailer. Secret bilateral contracting thus extends the opportunism problem across manufacturers, and it can rule out collusion even when the usual repeated-game condition would otherwise be satisfied.

The conclusion changes when retailers correctly anticipate the manufacturers' collusive strategies. Manufacturers can support collusion with grim-trigger strategies: a deviation from the collusive contracts triggers a permanent reversion to competitive wholesale prices. In this case, industry-profit-maximizing collusion can be sustained if manufacturers are sufficiently patient. Anticipated punishment is effective because retailers expect the rival to receive competitive wholesale offers after a deviation. Retailers do not accept supra-competitive

wholesale prices, so manufacturers cannot earn collusive profits once punishment begins. The stability condition is then the same as the condition faced by vertically integrated manufacturer--retailer pairs.

Full anticipation of collusive strategies, however, gives rise to an additional problem. Even if neither manufacturer wishes to deviate from the agreement unilaterally, the manufacturers may jointly prefer to revise their contracts taking the retailers' expectations as given. We call the absence of such a profitable joint deviation opportunism-proofness. This condition is more demanding than the conventional stability condition because a joint deviation exploits the retailers' beliefs: their expectations remain those of the collusive path, and the manufacturers can profit from revising both contracts against them.

We combine opportunism-proofness with the requirement that manufacturers do not renegotiate the wholesale price during the punishment phase. Together, these conditions adapt the concept of renegotiation-proofness developed by Farrell and Maskin (1989) to upstream collusion. We show that secret contracting with retailers can support a renegotiation-proof collusive equilibrium. Here, retailer expectations make punishment credible: after a deviation, retailers expect competitive terms for their rival and refuse the original collusive contracts. This prevents manufacturers from restoring collusion by mutual agreement, unlike vertically integrated firms, which can simply agree to resume collusion.

The expectations that make an established cartel stable and renegotiation-proof need not, however, allow it to form. If retailers anticipate that manufacturers use grim-trigger strategies, a history containing competitive conduct leads retailers to expect competitive conduct thereafter. Manufacturers may therefore be able to sustain collusion if retailers already believe that the industry is on the collusive path, while being unable to induce that belief after a competitive history. Hence, perfect anticipation solves the stability problem but can make the transition to collusion impossible.

We next consider adaptive beliefs, under which retailers adapt their assessment of whether manufacturers are colluding after observing a sequence of common high wholesale-price offers. They need not perfectly anticipate manufacturers' actions, but they can learn from observing repeated conduct that a broader change in the market regime has occurred. Adaptive beliefs generate a trade-off between cartel formation and opportunism-proofness. Faster adaptation reassures retailers more quickly that comparable terms apply to their rivals and therefore facilitates cartel formation. At the same time, it makes opportunistic joint deviations more attractive because retailers can be persuaded to return to the collusive outcome after fewer periods. Slower adaptation disciplines such opportunistic conduct but makes the transition from competition to collusion longer and more costly.

Adaptive beliefs also create a problem for conventional punishment strategies. Whenever they allow collusion to form after a competitive history, a punishment phase based on temporary reversion to competition is not renegotiation-proof: rather than completing the prescribed punishment, manufacturers would jointly prefer to start rebuilding the collusive regime immediately.

A particularly simple strategy illustrates how retailers' beliefs can substitute for conven-

tional cartel punishment. Suppose manufacturers always return to the collusive wholesale price after a deviation rather than prescribing a separate punishment phase. A deviation nevertheless changes retailers' beliefs and forces the manufacturers to rebuild the collusive regime. If the resulting loss from having to re-establish retailer confidence is sufficiently large, this belief-reset cost can discipline both unilateral and joint deviations. Thus, explicit punishment by cartel members is not necessary for collusion to be stable: retailer responses to a deviation can themselves provide discipline.

The analysis also has implications for competition policy. Vertical communication, resale price maintenance, recommended retail prices, and hub-and-spoke arrangements may have legitimate efficiency rationales, and our results do not imply that such practices are anticompetitive in themselves. The model identifies an additional channel through which they may facilitate upstream collusion: they can influence retailers' beliefs by signaling that higher wholesale prices apply throughout the industry. Competition authorities investigating a suspected supply-chain cartel should therefore ask not only whether upstream firms could monitor and punish one another, but also how they persuaded retailers that competing retailers faced comparable input-price increases. Communications or restraints that provide such reassurance may be relevant to the formation of a cartel even when retailers do not themselves take part in horizontal coordination.

The remainder of the paper is structured as follows. Section 2 relates our model to the literature. We set up the model in Section 3. Section 4 introduces the concept of renegotiation-proofness; it subsequently analyzes beliefs that are independent of the history of the game and beliefs that anticipate manufacturers' grim-trigger strategies. In Section 5, we study the formation of collusion and discuss beliefs that support this notion. We provide a competition-policy discussion in Section 6, where we relate our theory to business practices that manufacturers may use to deal with the problem of formation and opportunism. Section 7 concludes. The extension of collusion with trigger strategies is in Appendix A.1. Symmetric beliefs are analyzed in Appendix A.2, and linear wholesale prices are analyzed in Appendix A.3. All proofs for the main-text results are in Appendix B.

2 Related Literature

Our article relates to four strands of the literature: manufacturer collusion, the opportunism problem of a monopolist, retailers' beliefs under secret contracting, and cartel formation.

Manufacturer collusion. Nocke and White (2007) and Normann (2009) were the first to analyze tacit collusion among manufacturers in vertical relationships. Both compare an industry without integration to one in which a manufacturer-retailer pair is integrated, under two-part tariffs and linear prices respectively, and find that integration makes collusion easier to sustain. Piccolo and Miklós-Thal (2012) obtain similar results when retailers hold full bargaining power. Under public contracts, Schinkel et al. (2008) show that manufacturers

who must deter private damage claims by retailers need low wholesale prices and possibly negative franchise fees. Piccolo and Reisinger (2011) and Reisinger and Thomes (2017) study vertical restraints and channel structure: exclusive territories soften punishment but also reduce deviation profits and thus tend to facilitate collusion, and a common retailer differs from independent exclusive retailers under both public and private contracts. A further strand shows that resale price maintenance can facilitate manufacturer collusion, either when retailers privately observe local demand or cost shocks (Jullien and Rey, 2007) or, absent any uncertainty, when retail bargaining power would otherwise rule out supra-competitive wholesale prices (Hunold and Muthers, 2024). These articles focus on the manufacturers’ incentive constraints and abstract from retailer beliefs under secret contracting. In practice, manufacturer cartels have needed additional vertical instruments to make wholesale-price increases work. The German coffee roasters sustained higher wholesale prices only once they coordinated a retail price increase with their retailers (Holler and Rickert, 2022), other manufacturer cartels agreed additional contractual clauses such as payment conditions (Fink et al., 2017), and the US sugar cartel relied on exclusive brokers who were blacklisted if they deviated (Genesove and Mullin, 2001).

Opportunism problem. Colluding manufacturers resemble an upstream monopolist that contracts secretly with competing retailers, in that they compete with themselves when making coordinated offers (Hart and Tirole, 1990; O’Brien and Shaffer, 1992; McAfee and Schwartz, 1994). In such a scenario, the upstream firm wants to set high prices but cannot commit to refrain from increasing bilateral surplus with one retailer, which the other retailer anticipates. Vertical mergers and restraints—exclusive dealing, non-discrimination clauses, industry-wide RPM—have been read as devices for restoring upstream market power (O’Brien and Shaffer, 1992; McAfee and Schwartz, 1994; Rey and Vergé, 2004). Gaudin (2019) shows that, in the framework of Hart and Tirole (1990), linear contracting can raise retail prices absent collusion, and Do and Miklós-Thal (2025) revisit the seminal results under sequential (re)contracting. The collusion literature has not studied this opportunism problem, and we complement the literature by introducing an opportunism-proofness condition that is related to the renegotiation-proofness of Farrell and Maskin (1989).

Beliefs. The literature on secret contracting between manufacturers and retailers emphasizes the role of retailers’ beliefs (Hart and Tirole, 1990; McAfee and Schwartz, 1994; Segal, 1999), mostly focusing on passive, symmetric, or wary beliefs in static settings. Passive beliefs treat unexpected offers as mistakes, symmetric beliefs correspond to a rule of thumb whereby identical principals are conjectured to make identical offers (Pagnozzi and Piccolo, 2012), and wary beliefs, under which a retailer expects the rival to receive the offer that maximizes the manufacturer’s profit, are used where passive beliefs are implausible or equilibria fail to exist (Rey and Vergé, 2004; Rey and Tirole, 2007; Miklós-Thal and Shaffer, 2016). Passive and symmetric beliefs have been tested experimentally (Martin et al., 2001; Zhang, 2021), and Aoyagi et al. (2024) find in repeated games that the same history can generate

different beliefs and the same belief different actions.

In repeated games, the notion of passive beliefs differs across articles, as the static definition cannot be applied directly. We are closest to Piccolo and Reisinger (2011), Liu and Thomes (2021), and Gilo and Yehezkel (2020). Piccolo and Reisinger (2011) consider perfect Bayesian equilibrium with passive beliefs that are correct both on and off the equilibrium path. Liu and Thomes (2021) use passive and symmetric beliefs and assume that retailers understand when punishment takes place, so that beliefs are consistent with the manufacturers’ strategy profile in every subgame. Such consistency does not follow from weak perfect Bayesian equilibrium, which restricts nothing at nodes that are never reached, and it presumes that retailers correctly foresee conduct in subgames they cannot have experienced. Because different manufacturer strategies can support similar equilibrium paths, equilibrium-path observations need not identify off-path behavior. We therefore model beliefs explicitly, as different understandings of how the market works. Gilo and Yehezkel (2020) mirror our question from the other side: they study the manufacturer’s role in retailer collusion under rational beliefs, with repeated retailer interaction at the heart of the analysis, whereas we study the retailers’ role in manufacturer collusion and abstract from retailer collusion by assuming short-lived retailers.

Cartel formation. The literature on cartel formation asks how prospective members initiate cartels: through enforceable contracts and the stability of cartel membership (Selten, 1973; d’Aspremont et al., 1983), stochastic opportunities to form a cartel (Harrington and Chang, 2009), heterogeneous capacities and umbrella pricing (Bos and Harrington, 2010), signaling and mutual beliefs among producers (Harrington and Zhao, 2012; Harrington, 2017), antitrust as a facilitating factor (Andersson and Wengström, 2007; Bos et al., 2013, 2015), quality differentiation (Bos et al., 2020), and strategic uncertainty absent communication (Blume and Heidhues, 2008). We ask a different question: whether firms that have already reached—possibly explicitly—a common understanding to collude can move a currently competitive industry to the collusive outcome. At the core of that transition is the response of retailers, who are not themselves cartel members.

3 Model

We study manufacturer collusion in an infinitely repeated stage game with time $t = 0, \dots, \infty$.¹ There are two symmetric manufacturers, M_A and M_B , that compete by selling horizontally differentiated products to their exclusive retailers R_A and R_B . Each manufacturer privately offers a two-part tariff to its exclusive retailer with a unit wholesale price w_i and a franchise fee F_i , with $i \in \{A, B\}$. The offer remains unobservable to the rival pair within the period.²

¹We use the terms manufacturer for upstream firms and retailer for downstream firms. Our arguments naturally extend to any stage in a supply chain.

²An extension to retailers serving the products of both manufacturers as common agents is interesting but not straightforward. In this case, a retailer would receive offers of both manufacturers but would remain

The manufacturers are long-lived and discount next-period profits with the common discount factor δ with $0 < \delta < 1$. The retailers have a discount factor of zero, such that they do not take future profits into account.³

3.1 Set-Up

Timing and information within a period. In each period, the following stage game unfolds:

1. Contract offers: Each manufacturer privately offers a two-part tariff (w_i, F_i) to its retailer.
2. Acceptance: Each retailer decides whether to accept the offer.
3. Retail pricing: Retailers simultaneously and non-cooperatively set their retail prices p_i , $i \in \{A, B\}$.
4. End-of-period disclosure: At the end of the period, all actions (offers, acceptance decisions, and retail prices) are publicly observed.

The manufacturers' contract offers and the retailers' contract acceptance decisions are private within the period. Hence, a retailer cannot observe the contract offered to the rival. Moreover, when competing in the downstream market, the retailers are unable to observe each other's input prices and, thus, form beliefs about their rival's contract. At the end of each period, all actions are publicly observed, such that all players know the complete history of the game. We assume this to focus on the secret-contracting problem between manufacturers and retailers and to avoid manufacturer collusion being directly hampered by the long-term unobservability of manufacturers' actions to each other. The assumption implies that the supgame is a game of complete information but unobservable actions (within periods).

Assumptions on costs, demand, and profits. All (fundamental) costs are zero. We assume that the outside option (opportunity cost) of each retailer is equal to zero. We consider general demand functions that satisfy the standard properties summarized in

Assumption 1. *Demand $D_i(p_i, p_{-i})$ (with $i \in \{A, B\}$)*

uninformed about the offers that the other retailer receives. This may lead to a variant of the classic opportunism problem (Hart and Tirole, 1990) and to non-existence of equilibria (Rey and Vergé, 2010). In general, with more than two manufacturers, retailers may interact with only a subset of the manufacturers, yielding a similar setting as the one we study with two pairs of exclusive manufacturers and retailers.

³This assumption ensures that the retailers cannot collude. One can similarly assume that the retailers are short-lived but one should keep in mind that the retailers observe the history of the game. Related literature assumes that retailers are short-lived or only live for one period (see, for example, Piccolo and Reisinger, 2011 and Jullien and Rey, 2007). Such an assumption rules out both hub-and-spoke collusion (Harrington, 2018) and vertical collusion (Gilo and Yehezkel, 2020), where retailers are part of the collusion. As such, our assumptions appear to be in line with the cartel cases mentioned in the Introduction.

- decreases in the own price p_i : $\partial D_i(p_i, p_{-i})/\partial p_i < 0$,
- increases in the other product's price p_{-i} : $\partial D_i(p_i, p_{-i})/\partial p_{-i} > 0$, and
- the own-price effect dominates the cross-price effect: $|\partial D_i(p_i, p_{-i})/\partial p_i| > \partial D_i(p_i, p_{-i})/\partial p_{-i}$.

To ensure that there exists a unique and stable equilibrium in the downstream market, we assume that the Hessian matrix of $D_i(p_i, p_{-i})$ has a negative and dominant main diagonal. This results in well-behaved retail profits that are twice differentiable and concave. Note that this also implies that the retailers' reactions behave normally, such that $\partial p_i(w_{it}, p_{-i})/\partial w_{it} > 0$.

Our assumptions on retailer profits mostly carry over to manufacturer profits because manufacturers internalize retailer profits using two-part tariffs. In some cases, however, manufacturers' true actions and retailers' beliefs differ in such a way off the equilibrium path that the behavior of the manufacturer profit function differs from that of the retailer profit function. In those cases, we analogously assume that manufacturers' profits are well behaved, such that optimal behavior can be derived from the respective first-order conditions. We comment on these cases below.

Equilibrium concept. We consider (pure-strategy) weak perfect Bayesian equilibria (weak PBE) and focus on symmetric equilibria.⁴ We use the formal definition of a weak perfect Bayesian equilibrium from Mas-Colell et al. (1995): A profile of strategies and a system of beliefs is a weak PBE in extensive form games if it has the following properties:

1. The strategy profile is sequentially rational for the given belief system (each player maximizes expected utility at each information set, given its beliefs).
2. The system of beliefs is derived from the strategy profile through Bayes' rule whenever possible.

This implies that the retailers' beliefs are consistent with equilibrium strategies on the equilibrium path. Specifically, each retailer forms a belief about the contract offer made to the rival and about the rival's retail-pricing response after accepting its own contract in the second stage. Following Rey and Vergé (2004), we focus on retailers that form beliefs about the resulting retail price of the rival, which is the payoff-relevant information that retailers are lacking.⁵ Throughout the analysis we focus on the case in which the manufacturers correctly anticipate the retailer beliefs.

Our equilibrium concept is intended to capture strategic uncertainty at the retail level when contracts are private. On the equilibrium path, retailers' beliefs are consistent with

⁴The game defined includes two groups of players, manufacturers and retailers. Within each group, the players are symmetric. The equilibria we study are strongly symmetric in the sense that the (symmetric) manufacturers use a common continuation strategy on and off the equilibrium path (Athey et al., 2004; Jullien and Rey, 2007). The retailers play symmetric equilibrium actions in the downstream market.

⁵See Section 3 in Rey and Vergé (2004).

Bayes' rule whenever possible. Off the equilibrium path, we do not require subgame perfection because retailers cannot learn or verify manufacturers' off-path strategies when deviations occur with zero probability. To ensure that a manufacturer's deviation does not mechanically reveal private information about the rival's contract, for most of the analysis we impose

Condition 1. After observing its own period- t offer, retailer i 's belief about its rival depends on $\mathcal{H}_t$, but not on (w_{it}, F_{it}) ("no-signaling-what-you-don't-know").

Condition 1 has an intuitive interpretation in our game: A deviation by a retailer's own manufacturer should not change the retailer's belief about the other manufacturer's contract offer in the same period. The condition, then, is equivalent to within-period passive beliefs. Across periods, by contrast, beliefs can respond to any past action.⁶

In particular, we denote the history $\mathcal{H}_t \equiv [(w_{A0}, w_{B0}); \dots; (w_{At-1}, w_{Bt-1})]$ as the history of wholesale prices up to period $t - 1$. In the following, we restrict retailers' beliefs to depend on the history of wholesale prices only; in particular, they do not condition on the rival retailer's past prices. For each retailer i , the belief is thus a function $p_{-it}^e(\mathcal{H}_t)$.⁷

Beliefs. Weak perfect Bayesian equilibrium leaves retailers' beliefs unrestricted off the equilibrium path. The literature has resolved these beliefs in different ways, from passive beliefs — under which a retailer's belief about its rival is unaffected by its own offer (Hart and Tirole, 1990; Rey and Vergé, 2004) — to trigger-type beliefs that anticipate the collusive strategy and its punishments (Reisinger and Thomes, 2017). Rather than relying on one such specification, we let beliefs follow a parameterized family that maintains passive beliefs within each period (Condition 1) and nests much of the passive-belief literature as special cases. Retailers' beliefs take the following reduced form. Within each period beliefs are passive (Condition 1), but across periods they evolve at the end of the period according to the public information available. We can describe these beliefs *as if* retailers tracked a hidden conduct regime $\theta_t \in \{C, N\}$, which is either collusive (C) or competitive (N). Retailers do not observe this regime but assess it from observed wholesale prices. Let the assessment $\lambda_t \in [0, 1]$ be the weight a retailer places on the collusive regime, where λ_0 is the initial

⁶Condition 1 is derived from the definition of the PBE that Fudenberg and Tirole (1991a,b) provide. This definition is suited for games of incomplete information with independent types. Because in our game of complete information, the private information of the retailers is generated by manufacturers' actions, the concept can only be applied analogously. Similarly to Pagnozzi and Piccolo (2012), we adapt Condition B(iii) ("no-signaling-what-you-don't-know") on p. 332 in Fudenberg and Tirole (1991a) to our game. Originally, the condition means that, in a signaling game, the actions of other players with independent types have no effect on beliefs about a player's type if this player acts the same. In our setting, because retailers act simultaneously, one retailer cannot observe the other retailer's action so that the condition translates into: a retailer cannot infer anything about a rival's current pricing from the current action of its own manufacturer.

The condition is also in the spirit of sequential equilibrium, where (out-of-equilibrium) beliefs are naturally passive within a period because deviations are the result of a random mistake ("tremble") and these trembles are uncorrelated across players. Note that we cannot apply sequential equilibrium because we have a continuous action space.

⁷This is without loss of generality, as the wholesale prices contain all relevant information. We formally prove the result in Lemma 1.

assessment of retailers about the regime on the path of play.⁸

The assessment of the regime evolves through two channels, each triggered by the observed wholesale price. When both manufacturers set $w_{it} = w^C$, where w^C denotes the industry-profit-maximizing wholesale price, the assessment moves from competition toward collusion at the recovery rate ρ .

When a deviation is observed—that is, at least one manufacturer sets $w_{it} \neq w^C$ —the assessment moves from collusion toward competition at the detection rate τ .

The recovery rate ρ can be interpreted as the rate at which retailers realize that manufacturers are colluding. The detection rate τ is the rate at which retailers detect a breakdown of collusion, which is perfect at $\tau = 1$.

Definition 1. The assessment λ_t evolves according to the law of motion

$$\lambda_{t+1} = \begin{cases} \rho \cdot 1 + (1 - \rho) \cdot \lambda_t & \text{if } w_{At} = w_{Bt} = w^C \text{ in period } t, \\ \tau \cdot 0 + (1 - \tau) \cdot \lambda_t & \text{otherwise.} \end{cases}$$

Retailer i 's belief about the rival's retail price is

$$p_{-it}^e = \begin{cases} p^C & \text{if } \lambda_t \geq \lambda^*, \\ p^P & \text{if } \lambda_t < \lambda^*, \end{cases}$$

with an exogenous threshold $\lambda^* \in (0, 1)$, and where p^C is the collusive and p^P the competitive retail price.

Certainty about collusion corresponds to $\lambda_t = 1$, whereas certainty about competition to $\lambda_t = 0$. In a weak PBE λ_0 follows from the equilibrium strategies, as beliefs must be correctly adjusted to the equilibrium strategies of manufacturers at the relevant history.

Discussion of Assumptions. Our information structure with secret contracting captures the first stylized fact from Section 1: a manufacturer cartel cannot simply choose a wholesale price; it must induce retailers to accept contracts without observing whether rival retailers receive comparable terms. Retailers set prices simultaneously to capture exactly this uncertainty.

To analyze the game, we use weak PBE in conjunction with Condition 1 to be explicit about our equilibrium definition because there is a lack of clarity in the literature about the definition of perfect Bayesian equilibrium (PBE). According to the definition in Fudenberg and Tirole (1991a), our equilibrium would be considered a PBE. Other definitions of PBE, e.g., in González-Díaz and Meléndez-Jiménez (2014), require subgame perfection, which we

⁸We do not model retailers as Bayesian off the equilibrium path and hence, make no claim that λ_t is a posterior probability. Our reduced form features a two-state process in the tradition of Board and Meyer-ter Vehn (2013). The parameter λ_0 reflects the equilibrium belief in the initial period and hence, it is endogenous to the candidate equilibrium that is analyzed.

do not want to generally impose. In our setting, subgame perfection would require that the retailers correctly anticipate the manufacturers’ actions in information sets that are reached with zero probability. We explicitly want to allow for retailers’ incorrect judgment in these nodes.

Our belief definition nests much of the passive-belief literature as special cases. The parameters governing belief adjustment provide us with a reduced-form way to capture the second and third stylized facts. A higher recovery rate corresponds to environments in which retailers more quickly interpret repeated high wholesale offers as evidence of an industry-wide regime, possibly because of vertical communication, public signals, or observable pricing restraints. A lower recovery rate captures environments in which retailers remain skeptical for longer. Thus, the same belief process links the practical problem of creating mutual understanding to the theoretical distinction between forming collusion and sustaining it.

The reduced-form belief process in Definition 1 does not encompass all belief refinements used in the secret-contracting literature. Appendix A discusses two important benchmark specifications that fall outside this class: trigger beliefs with finite punishment, which require an explicit punishment state, and symmetric beliefs, which condition on the retailer’s current own offer and therefore violate Condition 1. The latter case could arise in a richer game, for instance, by including a cheap-talk stage. If retailers consider it possible that manufacturers may communicate, manufacturers’ actions might be correlated when a deviation occurs. Given this interpretation, Condition 1 does not naturally capture the spirit of sequential equilibrium in this “extended game.”

4 Sustainability of Established Collusion

We focus on the phenomenon of collusion among manufacturers and aim to address additional issues beyond the standard stability and incentive compatibility problem. In particular, we highlight the robustness of collusion to the opportunity of manufacturers to renegotiate, both during the collusive phase and after a deviation. Moreover, we shed light on the possibility for manufacturers to form collusion, that is, the transition from a competitive to a collusive market outcome. Both issues are theoretically and empirically relevant because collusion in real-world markets has various important aspects besides the stability problem. To characterize whether the collusive equilibrium allows for renegotiation or cartel formation, we define conditions that can be checked in a similar fashion to the traditional stability condition.

4.1 Preliminaries

First, we define renegotiation-proofness. Specifically, we rely on the notion of renegotiation-proofness by Farrell and Maskin (1989). We adapt the concept by restricting renegotiation to manufacturers. Renegotiation takes place implicitly in our model whenever *manufacturers* have a collective interest in revising their agreement. Formally, we hold retailers’ beliefs and

strategies constant when checking whether manufacturers have a Pareto-improving coordinated deviation. This assumption is in line with other theories on renegotiation-proofness (see, for example, Bernheim et al., 1987 and Bernheim and Ray, 1989).⁹

Definition 2. A collusive weak PBE $\mathcal{C}$, consisting of a strategy profile and a system of beliefs, is *weakly* renegotiation-proof if it features opportunism-proofness and renegotiation-proof punishment:

1. (Opportunism-Proofness) A collusive equilibrium strategy for given beliefs (a weak PBE) is said to be opportunism-proof if the manufacturers, on the equilibrium path, do not benefit from jointly changing their contract offers.
2. (Renegotiation-Proof Punishment) A weak PBE has the property of renegotiation-proof punishment if there is no punishment period in which the manufacturers would benefit if they jointly changed their contract offers.

The first condition excludes the scope for opportunistic manufacturer behavior during the collusive phase. Because demand does not change over time, manufacturers have no incentive to renegotiate the collusive price on the equilibrium path, but a coordinated one-shot deviation (against their retailers) may be profitable. Without opportunism-proofness, the implicit “collusive agreement” between the manufacturers is not renegotiation-proof. Whereas opportunism-proofness considers renegotiation-proofness of the collusive strategy on the collusive path, the criterion of renegotiation-proof punishment does so for the punishment phase after deviations. Intuitively, the condition states whether manufacturers want to negotiate the terms of the price war that follows after a unilateral deviation. Together, opportunism-proofness and renegotiation-proof punishment ensure *weak* renegotiation-proofness of the equilibrium analogously to the definition in Farrell and Maskin (1989). *Strong* renegotiation-proofness requires that not only all the continuation game equilibria do not invite joint deviation, but also that there is no other renegotiation-proof strategy profile that Pareto-dominates the candidate equilibrium of the whole game.

Definition 3. A collusive weak PBE $\mathcal{C}$, consisting of a strategy profile and a system of beliefs, is *strongly* renegotiation-proof if it is weakly renegotiation-proof and there exists no other renegotiation-proof strategy profile that Pareto-dominates $\mathcal{C}$.

Discussion. A collusive strategy profile is often viewed as all or nothing and non-adjustable. We show that manufacturers might want to collectively adjust their collusive strategies in a

⁹Our approach is in line with the original idea presented by Farrell and Maskin (1989) who consider subgame perfect Nash equilibria (SPNE) in a collusive game and treat the buyers as non-strategic. Similarly, renegotiation-proofness is usually applied to the parties of a *relational contract* only, in our case the collusive agreement, e.g., Buehler and Gärtner (2013); Goldlücke and Kranz (2013). Alternatively, one can understand our approach as applying the exact concept of Farrell and Maskin (1989) to the reduced form SPNE, taking retailers’ strategies and beliefs from the weak PBE as fixed.

stable market environment to the detriment of their retailers, which we refer to as opportunism. Additionally, we check whether manufacturers can and want to renegotiate in the punishment phase after a deviation. As stated by Levenstein and Suslow (2006), “one of the most clearly established stylized facts is that cartels form, endure for a period, appear to break down, and then re-form again”. They report that cheating and disagreement in cartels happen quite often but cartels frequently have multiple episodes of cooperation. Hence, it might be that cartel members renegotiate once cooperation among them has failed to return to the collusive outcome. As discussed by Levenstein and Suslow (2006), empiricists are able to differentiate between bargaining price wars and price wars that are part of a punishment strategy. We will show that collusion can be sustainable either with or without renegotiation-proofness, explaining both kinds of price wars.

4.2 Non-Reactive Beliefs

In this section, we establish benchmark results by assuming that the retailer beliefs do not react to past actions in the game. Such beliefs arise naturally when the manufacturers’ equilibrium strategies are stationary. For example, the competitive equilibrium in which both manufacturers set the competitive wholesale prices in each period is consistent with retailers having a constant belief about wholesale prices. With these non-reactive beliefs, we show that collusion is infeasible.

Remark 1. (Non-reactive beliefs: $\rho = 0, \tau = 0$) Retailer i ’s expectation p_{-it}^e about its rival’s retail price is independent of the public history and of its own current offer (w_{it}, F_{it}) .

With $\tau = 0$ retailers stick to their initial assessment in cases of non-collusive prices and with $\rho = 0$ the beliefs cannot move towards believing in collusion from competition in the case of collusive prices. The assessment of the regime is hence constant at $\lambda_t = \lambda_0$ in all periods. This belief refinement defines the retailers’ out-of-equilibrium beliefs about the retail price of their competitor to be independent of past actions, while beliefs remain passive within the period in line with Condition 1. The next lemma shows that it is without loss to focus on beliefs over retail prices and that it is without loss that those beliefs condition on the history of wholesale prices only. The intuition behind this result is that retail prices are the payoff-relevant information for retailers, while the history of actions of manufacturers, captured by their wholesale price offers, contains the relevant information about the past.

Lemma 1. *Let Condition 1 hold. Suppose that players use pure strategies and use undominated contract offers. Then, it holds that (i) for any wholesale price w_{it} and any belief p_{-it}^e , retailer i ’s price is uniquely determined by $p_i(w_{it}, p_{-it}^e)$ and is independent of F_{it} , (ii) any belief w_{-it}^e about the rival’s wholesale price implies a unique belief p_{-it}^e about the rival’s retail price, and (iii) it is without loss of generality to restrict retailers’ beliefs to depend on $\mathcal{H}_t$.*

Proof. See Appendix B. □

We continue by solving the game. We first solve for the equilibrium of the game under competition that results when both manufacturers maximize stage-game profits. In the last

stage within each period, on the equilibrium path, each retailer has accepted the contract, but the rival's wholesale price remains secret. Each retailer i observes its own wholesale price w_{it} and holds a non-reactive belief p_{-it}^e about the retail price of the rival. The retailers set the retail prices p_{it} simultaneously. Thus, the profit of retailer i is

$$\pi_{it}(p_{it}, p_{-it}^e) = (p_{it} - w_{it}) D_i(p_{it}, p_{-it}^e) - F_{it}.$$

Retailer i maximizes its profit with respect to p_{it} . The first-order condition is

$$D_i(p_{it}, p_{-it}^e) + (p_{it} - w_{it}) \frac{\partial D_i(p_{it}, p_{-it}^e)}{\partial p_{it}} = 0 \quad (1)$$

and defines retailer i 's reaction function $p_i(w_{it}, p_{-it}^e)$. Anticipating the pricing outcome and resulting profits, each retailer decides whether to accept the wholesale tariff offered by its manufacturer in the second stage. Under Assumption 1, there exists a unique symmetric Nash equilibrium in the downstream market for offers (w_{it}, F_{it}) made by the manufacturers conditional on retailers accepting the offers.

In the first stage, the manufacturers offer their wholesale tariffs. Under manufacturer competition, each manufacturer i maximizes its profit, anticipating that its retailer sets a price of $p_i(w_{it}, p_{-it}^e)$:

$$\max_{w_{it}, F_{it}} w_{it} \cdot D_i(p_i(w_{it}, p_{-it}^e), p_{-i}(w_{-it}, p_{it}^e)) + F_{it},$$

subject to the retailer's participation constraint

$$(p_i(w_{it}, p_{-it}^e) - w_{it}) \cdot D_i(p_i(w_{it}, p_{-it}^e), p_{-it}^e) - F_{it} \geq 0. \quad (2)$$

The profit is determined by two parts. The variable part consists of the units sold times the unit wholesale price, and the fixed part consists of an up-front payment from the retailer to the manufacturer. The maximum fixed payment cannot be larger than the retailer's operating profit before the fixed fee. This participation constraint binds in equilibrium.

In the one-shot equilibrium, the belief of the retailer about the retail price of the rival coincides with the expectation of the manufacturer about the retail prices. This implies $p_{-i}(w_{-it}, p_{it}^e) = p_{-it}^e$.

The optimality condition of the manufacturer problem implies that the only solution has $w_{it} = 0$. Denote the competitive wholesale price by w^P and the resulting retail price by $p^P = p_i(0, p^P)$. We have the following result:

Lemma 2. *With non-reactive beliefs, in the equilibrium with competing manufacturers, the wholesale prices are equal to zero ($w^P = 0$), resulting in the competitive retail price p^P .*

Proof. See Appendix B. □

The intuition behind the result is that, by setting the wholesale price equal to marginal

costs, each manufacturer ensures that retailers maximize the joint profits of a manufacturer-retailer pair, which can then be extracted via the fixed fee.

This non-reactive benchmark defines the competitive wholesale price $w_i = 0$, the corresponding fixed fees, and the resulting retail prices. The fixed fee is equal to the retailer's profit, $F_i = \pi_i(p^P, p^P)$, corresponding to a non-colluding industry. In the next steps, we characterize the equilibria of the repeated game if retailers hold non-reactive beliefs. To show that firms are unable to collude on a price different from the competitive wholesale price of zero, we take the whole dynamic game into account. Manufacturers may collude using any dynamic strategy (for example, grim-trigger strategies). Each strategy, however, is a mapping from the previous history to an action chosen in period t .

Proposition 1. *Suppose retailers hold non-reactive beliefs. Then, there exists a unique symmetric weak perfect Bayesian equilibrium in which the manufacturers set wholesale prices equal to zero and extract all profits of retailers via the fixed fee. Collusion cannot increase prices above the competitive level.*

Proof. See Appendix B. □

To understand this impossibility result, consider that retailers might hold a non-reactive belief that is consistent with collusion. However, if a manufacturer deviates, these beliefs do not react. Each manufacturer can therefore secure a profit of $\Pi(0, p_{it}^e)$ in every period by setting a wholesale price of zero, irrespective of its rival's action, and this profit strictly exceeds what any candidate collusive contract yields. Manufacturers are thus insulated from punishment and the incentive to deviate always dominates. This logic eliminates any collusive equilibrium above the competitive price level. The strict impossibility result depends on efficient contracts (two-part tariffs are sufficient here).¹⁰ This result formalizes the implementation problem described in the first stylized fact. Even if manufacturers would like to collude, a retailer that does not expect the rival retailer to face a comparable wholesale increase will not behave in a way that supports the collusive outcome. In this sense, skeptical, or stubborn, retailer beliefs can be self-fulfilling. Note that the proof of Proposition 1 extends beyond the parametrization of beliefs that we assume here to any retailer belief that does not react to the history of actions and thus also to deterministic belief paths that may vary with time.

4.3 Grim-Trigger Beliefs

We turn to belief specifications under which retailers fully anticipate the manufacturers' strategies, including both the collusive path and the associated punishment following a deviation. For this, we consider that manufacturers play grim-trigger strategies. The respective

¹⁰The classical opportunism problem of an upstream monopolist can be mitigated under linear contracts as shown by Gaudin (2019), where the monopolist achieves supra-competitive (marginal) wholesale prices. In contrast, with two-part tariffs, the monopolist sets the wholesale prices (w) equal to its marginal costs (McAfee and Schwartz, 1994). We provide a result for linear tariffs in Section A.3.

beliefs correspond to what is implicitly assumed in much of the repeated-game literature on collusion, where continuation play is understood to follow grim-trigger strategies and retailers correctly infer whether the industry is in a collusive or punishment phase. In this sense, grim-trigger beliefs represent the benchmark case closest to subgame-perfect reasoning in environments with private contracts. The grim-trigger-belief benchmark can therefore be interpreted as the limiting case in which the second stylized fact is fully operative: retailers understand not only that manufacturers collude, but also how the collusive arrangement responds to deviations. Vertical communication or restraints may move real-world environments closer to this benchmark by reducing retailer skepticism.

Collusive strategy. Manufacturers play grim-trigger strategies: manufacturers set the collusive wholesale price w^C unless one manufacturer deviates. Any deviation by one manufacturer causes the other manufacturer to set the punishment price w^P forever. The punishment with w^P must be individually rational, which implies that $w^P = 0$ must hold in equilibrium. This follows from the logic presented in the proof of Proposition 1: Otherwise, each manufacturer would have an incentive to deviate from a $w^P \neq 0$ in a punishment period in which future beliefs and actions are fixed and unaffected by current actions. In punishment periods, each manufacturer must thus maximize short-term profits. We focus on industry-profit-maximizing collusion, such that $w^C = w^{M11}$, i.e., collusion at the prices that maximize the integrated industry profit:

$$p^M = \arg \max_p p \cdot D_i(p, p) + p \cdot D_{-i}(p, p). \quad (3)$$

The above equation determines the collusive retail price p^M , and the corresponding wholesale price w^M is implicitly defined through $p^M = p_i(w^M, p^M)$. Denote the resulting maximal industry profit by

$$\Pi^M := p^M \cdot D_i(p^M, p^M) + p^M \cdot D_{-i}(p^M, p^M). \quad (4)$$

Thus, whenever the manufacturers collude at the industry-profit-maximizing wholesale price w^M , each manufacturer earns a profit of

$$\Pi^C := \frac{\Pi^M}{2}.$$

The optimal collusive wholesale price maximizes the joint profit of the manufacturers given that retailers expect the rival retailer to charge the corresponding collusive retail price p^M . The collusive profits are Pareto-optimal for the manufacturers if there is no joint deviation that is more profitable (see Definition 2). If weak renegotiation-proofness holds, then the equilibrium is also strongly renegotiation-proof because the profit on the equilibrium path would be Pareto-efficient for the colluding manufacturers.

¹¹Qualitatively, our results do not rely on the assumption that manufacturers collude on the monopoly price level. Collusion at the industry-profit maximizing level, however, facilitates renegotiation-proofness.

Beliefs. We define grim-trigger beliefs as beliefs that fully anticipate the grim-trigger strategies of manufacturers. In our parameterization, this means that retailers do not revert their assessment to collusion once a non-collusive wholesale price has been observed, which is captured by $\rho = 0$ (grim). Instantaneous and complete adjustment once a deviation is observed is captured by $\tau = 1$ (trigger). We focus on the candidate collusive equilibrium with industry-maximizing prices and punishment price p^P being equal to the competitive price. Our definition of beliefs then implies the following remark.

Remark 2. (Grim-trigger beliefs: $\rho = 0, \tau = 1$) For the candidate collusive equilibrium, we set $\lambda_0 = 1$, that is, retailers assign full weight to the collusive regime initially. Therefore, retailers believe that the rival sets a retail price of p^C in the first period. As long as both manufacturers play w^C , each retailer believes that the other retailer sets p^C in the current period. Once one manufacturer has deviated in the previous period, both retailers believe that the other retailer sets the competitive price p^P .

Given the passive nature of beliefs within a period (Condition 1), the beliefs are not correct in deviation periods. Neither a retailer that is offered the equilibrium contract updates its belief about the other retailer's contract offer, nor does the retailer that receives a deviating offer update its belief about the rival's contract offer. Because a deviation does not occur on the equilibrium path, the beliefs are nevertheless correct on the equilibrium path. Off the equilibrium path, the retailers' beliefs anticipate that manufacturers play grim-trigger strategies in that they also punish a deviation by one manufacturer in period t in all future periods.

Equilibrium. To determine an equilibrium of the dynamic game, we must consider deviations from the collusive strategy. In equilibrium, each manufacturer realizes per-period profits of $\Pi^C = \Pi^M/2$. In a deviation period, both retailers believe that the wholesale price is w^C and anticipate that the other retailer sets p^C . Retailer i therefore sets $p_{it} = p_i(w_{it}, p^C)$. Suppose that manufacturer i maximizes its deviation profit in period t . When there is a deviation, that is, $w_{it} \neq w^C$ holds, grim-trigger beliefs imply that the level of w_{it} has no impact on future beliefs. The deviation profit is given by

$$\begin{aligned} & w_{it} \cdot D_i(p_i(w_{it}, p^C), p^C) + [p_i(w_{it}, p^C) - w_{it}] \cdot D_i(p_i(w_{it}, p^C), p^C) \\ & = p_i(w_{it}, p^C) \cdot D_i(p_i(w_{it}, p^C), p^C). \end{aligned} \quad (5)$$

Maximizing with respect to w_{it} yields the first-order condition

$$\begin{aligned} & \frac{\partial p_i(w_{it}, p^C)}{\partial w_{it}} D_i(p_i(w_{it}, p^C), p^C) + \frac{\partial D_i(p_i(w_{it}, p^C), p^C)}{\partial p_i} \frac{\partial p_i(w_{it}, p^C)}{\partial w_{it}} p_i(w_{it}, p^C) = 0 \\ & \iff \frac{\partial p_i(w_{it}, p^C)}{\partial w_{it}} \frac{\partial D_{it}(p_i(w_{it}, p^C), p^C)}{\partial p_{it}} w_{it} = 0. \end{aligned} \quad (6)$$

The last step follows from the first-order condition of the retailers in equation (1). Inserting the expression in the first line above yields the second line. Because the first two factors in equation (6) are non-zero, the manufacturer optimally deviates to $w^D = 0$. This results in

$$\Pi^D := p_i(0, p^C) \cdot D_i(p_i(0, p^C), p^C).$$

After any deviation by a manufacturer, the beliefs revert to $p_{-i}^e = p^P$ forever, that is, the belief in the punishment period. This results in profits of

$$\Pi^P := p^P \cdot D_i(p^P, p^P), \quad (7)$$

where $p^P = p_i(0, p^P)$ is the competitive price.

Collusion is sustainable when no manufacturer wants to deviate from the grim-trigger strategy. Using the one-shot deviation principle, the relevant incentive constraint for stability is

$$\frac{\Pi^C}{1 - \delta} \geq \Pi^D + \frac{\delta \Pi^P}{1 - \delta}. \quad (8)$$

The left-hand side contains the present value on the equilibrium path and the right-hand side the present value of a deviation. We can rewrite the incentive constraint for manufacturer i as follows:

$$\frac{p^C \cdot D_i(p^C, p^C)}{(1 - \delta)} \geq p_i(0, p^C) \cdot D_i(p_i(0, p^C), p^C) + \frac{\delta}{1 - \delta} (p^P \cdot D_i(p^P, p^P)). \quad (9)$$

This inequality is equivalent to the incentive constraint for standard horizontal collusion (when manufacturers and retailers are pairwise integrated), where the following order holds: $\Pi^D > \Pi^C = \Pi^M/2 > \Pi^P$.

Let us check whether the equilibrium is opportunism-proof (see Definition 2). Jointly, the manufacturers may have an incentive to reduce their wholesale prices for given beliefs. Suppose $\mathcal{H}_C$ is the history of the game, such that each retailer believes that its competitor sets p^C and anticipates setting $p_i(w_{it}, p^C)$. First, we show that jointly deviating manufacturers set a wholesale price of $w_{it} = w_{-it} < 0$ to maximize spot profits. To see this, let us inspect the profit per manufacturer in the case of a joint deviation:

$$\begin{aligned} \Pi^{JD} := & \frac{1}{2} \max_{w_{it}, w_{-it}} w_{it} \cdot D_i(p_i(w_{it}, p^C), p_{-i}(w_{-it}, p^C)) + [p_i(w_{it}, p^C) - w_{it}] \cdot D_i(p_i(w_{it}, p^C), p^C) \\ & + w_{-it} \cdot D_{-i}(p_{-i}(w_{-it}, p^C), p_i(w_{it}, p^C)) + [p_{-i}(w_{-it}, p^C) - w_{-it}] \cdot D_{-i}(p_{-i}(w_{-it}, p^C), p^C). \end{aligned}$$

We assume that this profit is quasi-concave, such that we can use first-order conditions.¹² Using the retailers' first-order conditions in equation (1) by applying symmetry: $w_{it} = w_{-it} = w$, we simplify the first-order conditions of the manufacturers' joint maximization problem. This is possible because we assume that the manufacturer profits are well behaved in the

¹²Note that this holds for linear demand. In general, quasi-concavity depends on higher-order derivatives of demand at different loci of demand.

sense that the optimal joint action of the manufacturers is symmetric. This yields

$$\left[\underbrace{D_i(p_i(w, p^C), p_{-i}(w, p^C)) - D_i(p_i(w, p^C), p^C)}_{<0} \right] + w \underbrace{\frac{\partial p_i}{\partial w_{it}}}_{>0} \left[\underbrace{\frac{\partial D_i}{\partial p_i} + \frac{\partial D_{-i}}{\partial p_i}}_{<0} \right] = 0, \forall i. \quad (10)$$

Equation (10) only holds for a wholesale price w that is below zero. Hence, manufacturers optimally deviate to this w jointly. A manufacturer makes a higher profit in the case of a joint deviation than when deviating unilaterally: $\Pi^{JD} > \Pi^D$. To explain the last inequality, note that the manufacturers could replicate the profit of Π^D for each of them by setting $w = 0$. The manufacturers, however, optimally set a price w below zero because this yields strictly larger profits. Setting a price below zero has a negative externality on the rival retailer that they do not internalize; it enables them to profitably exploit the incorrect beliefs. Following a deviation, the manufacturers make profits Π^P in future periods due to the grim-trigger beliefs. This results in the following opportunism-proofness condition

$$\frac{\Pi^C}{1 - \delta} \geq \Pi^{JD} + \frac{\delta \Pi^P}{1 - \delta}. \quad (11)$$

Comparing equation (11) with equation (8), the only difference is the deviation profit on the right-hand side. Because $\Pi^{JD} > \Pi^D$, the condition for opportunism-proofness is harder to satisfy than the stability condition (9).

Proposition 2. *With grim-trigger beliefs, there exists an equilibrium in which manufacturers are able to sustain collusion on the industry-profit-maximizing wholesale price using grim-trigger strategies if the discount factor is high enough to satisfy Equation (9). This condition is equivalent to the incentive constraint when manufacturers and retailers are pairwise integrated, that is, under horizontal collusion. Under the stronger condition (11), collusion is also opportunism-proof. The punishment is renegotiation-proof.*

Proof. See Appendix B. □

First, in contrast to Proposition 1, the manufacturers are able to sustain collusion when the retailers hold grim-trigger beliefs. Intuitively, grim-trigger beliefs allow for punishment, which is ineffective with non-reactive beliefs. Moreover, the equilibrium above satisfies subgame perfection, that is, retailers correctly anticipate the manufacturers' actions in information sets that are reached with zero probability. The retailers have consistent beliefs on the equilibrium path in the collusive phase and in punishment phase subgames, as grim-trigger beliefs mirror the manufacturer's grim-trigger strategies.

Second, as derived above, the opportunism problem gives rise to the incentive condition (11) that is harder to satisfy (that is, only for more patient firms) than condition (8) for stable collusion. Hence, opportunism can make cartels less sustainable. Similar to the literature on the opportunism problem in single-shot games, this result depends on the beliefs. Due to their passive beliefs, the retailers do not react immediately to opportunism, which allows the

manufacturers to “trick” the retailers because the price that retailers expect will turn out to be incorrect if manufacturers jointly deviate. This joint deviation differs from a unilateral deviation of one manufacturer-retailer pair from a candidate equilibrium in which instead the belief of the deviating retailer is correct. In the latter case, it is optimal for the deviating manufacturer to set the wholesale price equal to the true costs. In the former case of a joint deviation, each retailer wrongly believes that the other retailer will buy at a high wholesale price and will thus sell at a high price, such that demand is high. It is, hence, profitable for the manufacturer to set a high fixed fee in return for a marginal wholesale price below costs because the retailer believes that it can sell a large quantity. A marginal wholesale price below costs, however, only becomes profitable for a manufacturer when the retailer has an incorrect belief. It could thus signal to the retailer that its belief is wrong because, if it were correct, the manufacturer’s offer would be dominated by an offer with a wholesale price of $w = 0$. The stricter condition for opportunism-proofness is thus an implication of the passive nature of the beliefs. We later consider symmetric beliefs, where retailers’ beliefs instantly react to any change in wholesale prices.

If marginal wholesale prices below marginal costs are impossible (for instance, due to competition law), the incentive constraints for opportunism and for stability are identical, implying that any stable collusive equilibrium is also opportunism-proof. In any case, the opportunism problem is mitigated under grim-trigger beliefs compared to the one-shot game, or non-reactive beliefs, because retailers with grim-trigger beliefs react to the deviation from the expected cartel wholesale prices by adapting their beliefs from the next period onward toward the competitive belief. This effectively punishes manufacturer opportunism, such that joint deviations are not profitable when manufacturers are sufficiently patient.

The characterized equilibrium is also renegotiation-proof in the sense of Farrell and Maskin (1989). Because the opportunism-proofness condition (11) ensures that manufacturers have no incentives to deviate jointly from the collusive agreement, and punishment is renegotiation-proof as well, the equilibrium is *weakly* renegotiation-proof. As the equilibrium features collusion on the industry-profit-maximizing prices that are Pareto-optimal for the manufacturers, the equilibrium is also *strongly* renegotiation-proof. This yields the next corollary.

Corollary 1. *The equilibrium with grim-trigger beliefs described in Proposition 2 is strongly renegotiation-proof if the opportunism-proofness condition (11) is satisfied.*

For reference, consider a standard model of horizontal collusion absent a vertical dimension. In our model, this arises when each manufacturer-retailer pair is vertically integrated. Suppose that firms collude with grim-trigger strategies at the industry-profit-maximizing level and the usual stability condition is met. Collusion is opportunism-proof because joint profits are maximized. In this industry structure, punishment is not renegotiation-proof because the firms would prefer to renegotiate to return to collusive prices. Hence, in the standard horizontal-collusion benchmark, the grim-trigger equilibrium with reversion to competitive prices is not renegotiation-proof.

By contrast, we will show that secret contracting can feature renegotiation-proof punishment and that opportunism-proofness can impose additional conditions on collusion.¹³

Empirically, this wedge suggests that upstream cartels operating through independent retailers should, other things being equal, exhibit more durable punishment phases than horizontal cartels selling directly to final consumers: once collusion breaks down, retailers who no longer accept collusive terms prevent the manufacturers from renegotiating their way back, whereas horizontally colluding firms face no such external obstacle to re-coordination. Breakdowns of vertically mediated collusion should therefore be followed by longer competitive spells — and re-formation, where it occurs, should coincide with vertical devices that rebuild retailer acceptance, such as resale price maintenance or manufacturer-led communication.

5 Forming Collusion

Next, we characterize how manufacturers can overcome the coordination problem by characterizing the conditions under which collusion is formable in the sense that collusion can be started and maintained. For example, suppose that manufacturers have been competing for a large number of periods and want to start collusion with trigger strategies. Formability addresses the question of whether the system of retailers' beliefs allows for a transition path to a collusive equilibrium. The formability condition thus formalizes the third stylized fact. It asks whether manufacturers can move from a non-collusive history to a collusive path when retailers initially interpret high wholesale offers through the lens of past competitive conduct. Thus, formation is not simply the first period of a stable cartel; it is a separate constraint generated by downstream beliefs.

5.1 Preliminaries

We define a weak PBE as formable if there is a transition path to a collusive equilibrium in the sense that, in this transition, each player maximizes its expected utility under the belief system of the weak PBE. Thus, formability is an additional property of an equilibrium that imposes a condition on the system of beliefs.

Definition 4. (Formable) A collusive weak PBE $\mathcal{C}$, consisting of a strategy profile and a system of beliefs, is formable if there exists a (transition) strategy profile, such that for the system of beliefs of $\mathcal{C}$ after an *arbitrary* history $\mathcal{H}_{t_0}$:

1. In the transition phase starting in period t_0 and lasting for x periods, with $x < \infty$,

¹³Recall that the concept of renegotiation-proofness is applied to the coalition of manufacturers; but the result may be more general if retailers are allowed to be part of the negotiation, as long as beliefs are not affected. Recall that we focus on collusion in which retailers do not take part. Fixing beliefs, however, manufacturers leave zero profits to retailers, making them effectively indifferent in renegotiation, such that a renegotiation-proof equilibrium would also be renegotiation-proof when including retailers.

until $t_0 + x$ (exclusively), the strategy profile in the continuation game is sequentially rational for the system of beliefs of C ;

2. The collusive phase starts in period $t_0 + x$. In that continuation game, the strategy profile of C , adjusted with $t_0 + x$ as the first period, together with the system of beliefs of C , is a weak PBE.

Note that beliefs in the transition phase can be incorrect, such that the strategy profile and system of beliefs after the alternative history do not generally constitute a weak PBE.¹⁴ The number of transition periods x can be long, such that our definition is not restrictive with regard to the speed of formation.

Discussion. In a standard model of horizontal collusion, the usual stability condition is necessary for collusion and also sufficient if the manufacturers can coordinate on the collusive outcome through a “meeting of the minds.” If, however, colluding upstream firms sell to competing downstream firms, stability is not sufficient. Retailers might not accept the new contract offers with collusive prices, especially if the contract offers to downstream rivals are secret. Our formation condition captures whether manufacturers can implement a collusive agreement vis-à-vis the downstream firms. For example, in the coffee cartel, collusion among the coffee roasters turned out to be unsuccessful because their retailers did not accept the collusive wholesale prices (Holler and Rickert, 2022). Collusion only became successful after introducing resale price maintenance.

5.2 Non-Reactive Beliefs and Grim-Trigger Beliefs

Before turning to beliefs that support formation, we show that the two benchmark specifications of Section 4 do not — though for different reasons. With non-reactive beliefs, the question of formability does not arise. Proposition 1 establishes that there is no collusive weak PBE to be formed: because beliefs never respond to observed conduct, a manufacturer’s profit in any period is insulated from its rival’s action, such that undercutting is optimal irrespective of the history and prices cannot exceed the competitive level.

Under grim-trigger beliefs, by contrast, a collusive equilibrium exists but cannot be reached. These beliefs are fully anticipatory and unforgiving: any deviation from collusive wholesale prices leads retailers to permanently believe that the industry is in the punishment phase. To see that collusion is *not formable*, consider any history in which non-collusive wholesale prices have previously been observed. With this history, grim-trigger beliefs imply that the belief is p^P forever. As a result, retailers never accept collusive contracts in subsequent periods. The second condition of formability therefore fails: there exists no finite transition after which the collusive equilibrium can be reached for arbitrary histories. Hence,

¹⁴The definition, however, requires that the strategy profile of the continuation game starting with t_0 is perception perfect, that is, sequentially rational given the beliefs.

beliefs that are closest to the standard subgame-perfect logic in repeated games support collusion only once it is already in place.

5.3 Example of Formable Beliefs: Adaptive Beliefs

We next introduce a class of adaptive beliefs to illustrate that formability requires belief systems that are sufficiently flexible to allow for a transition from competition to collusion. For our parametrization of beliefs this implies that $\rho > 0$ is required to move from an assessment of non-collusion to an assessment of collusion, which is a necessary condition for formability. We do not provide a complete theory of belief formation. Rather, the adaptive beliefs provide a reduced-form representation of retailer learning that is governed by the information structure of the model and by minimal consistency requirements. This approach captures a minimal form of heuristic learning under strategic uncertainty. It allows beliefs to switch between competitive and collusive regimes, which is necessary for formability, while remaining consistent with the benchmark case of fully anticipatory (grim-trigger) beliefs when adaptation is arbitrarily slow. We do not claim that adaptive beliefs are unique or optimal; they demonstrate that collusion can be formable once beliefs are allowed to respond to patterns in observed behavior.

Next, we formally introduce adaptive beliefs to analyze equilibria with (grim-)trigger strategies. From our parameterization it follows that for $\rho > 0$ and $\lambda_0 < \lambda^*$ there is a finite number of periods T , after which beliefs adapt to a change in manufacturer conduct.

Remark 3. (Adaptive Beliefs: $\rho \in (0, 1], \tau = 1$) For $0 < \rho < 1$, the assessment rises geometrically: $1 - \lambda_t = (1 - \lambda_0)(1 - \rho)^t$ and reaches the threshold after observing $w_A = w_B = w^C$ for $T(\rho, \lambda^*) = \left\lceil \frac{\log(1 - \lambda^*) - \log(1 - \lambda_0)}{\log(1 - \rho)} \right\rceil$ periods. For $\rho = 1$ and $\lambda_0 < \lambda^*$, $T = 1$. For any $\lambda_0 \geq \lambda^*$, the assessment falls to zero in one period after observing a $w_i \neq w^C$. Note that T is (weakly) decreasing in ρ and (weakly) increasing in λ^* .

The definition of adaptive beliefs differs from the definition of (grim-) trigger beliefs in that we allow retailers to adopt the collusive belief after observing w^C for sufficiently many periods. In the following, we use grim-trigger strategies to analyze the necessary incentive constraints implied by opportunism-proofness, renegotiation-proof punishment, and formability of collusion that arise from considering adaptive beliefs.

Stability. The stability condition on the collusive equilibrium path, for $\lambda_0 = 1$, is

$$\frac{\Pi^C}{1 - \delta} \geq \Pi^D + \frac{\delta \Pi^P}{1 - \delta}. \quad (12)$$

In this case, retailers with adaptive beliefs already have the belief p^C and revert to the belief p^P after a deviation. Hence, with grim-trigger strategies, the condition is the same as the incentive condition (8). As a consequence, collusion is stable if manufacturers are sufficiently patient (δ is large enough). Again, the condition is identical to the stability condition under horizontal collusion.

Formability. Formability holds if there exists a transition path to the collusive weak PBE, where – on the path – the manufacturers’ actions are mutually best responses. Suppose that, starting from any history in period s , the manufacturers both start to play the collusive wholesale price w^C with the usual grim-trigger strategies that punish any deviation. If the manufacturers want to form collusion, the worst history in terms of our beliefs is that there was competition in $s - 1$ (with profits Π^P and $\lambda_s = 0$). This implies that, for the next T periods including period s , the retailers’ beliefs are fixed at p^P , such that the manufacturers can extract lower transfers. Throughout, let $p^F \equiv p_i(w^C, p^P)$ denote the retail price in a formation period, where $p^P < p^F < p^M$. Because the fixed fee is set against the belief p^P while both retailers charge p^F , the per-period manufacturer profit during formation is

$$\Pi^F = w^C D_i(p^F, p^F) + (p^F - w^C) D_i(p^F, p^P).$$

With grim-trigger strategies, manufacturers have an incentive to jointly start to collude in a competitive period, such that they eventually arrive at the collusive equilibrium path if

$$\frac{\Pi^P}{1 - \delta} \leq \frac{1 - \delta^T}{1 - \delta} \Pi^F + \frac{\delta^T}{1 - \delta} \Pi^C. \quad (13)$$

The left-hand side contains the present value of perpetual competition. The first term on the right-hand-side is the discounted profit of the T periods in which retailers’ belief is p^P , while manufacturers actually set w^C ; the second term is the discounted profit of perpetual collusion starting after T formation periods. Note that condition (13) may even hold for any level of patience. Manufacturers extract transfers against the retailers’ punishment belief p^P , but realized demand is higher because the rival retailer also raises its price, so a formation period need not be less profitable than a competitive one. Whenever $\Pi^F \geq \Pi^P$, the manufacturers’ joint incentive to begin forming collusion is immediate.

Note that against the belief p^P , w^C is not the manufacturers’ best response: each of them could raise its current profit to $\Pi^{F,D} > \Pi^F$ by lowering its wholesale price. Each manufacturer therefore has an incentive to deviate during formation. Deviating during the formation phase yields a period profit of

$$\Pi^{F,D} = \max_{w_i} w_i \cdot D_i(p_i(w_i, p^P), p^F) + [p_i(w_i, p^P) - w_i] \cdot D_i(p_i(w_i, p^P), p^P), \quad (14)$$

but triggers a punitive action forever. Consider the incentives to stick to w^C in the formation phase. No manufacturer wants to deviate unilaterally in the formation phase, which implies that actions in the transition are mutually best responses, if

$$\frac{1 - \delta^T}{1 - \delta} \Pi^F + \frac{\delta^T}{1 - \delta} \Pi^C \geq \Pi^{F,D} + \frac{\delta \Pi^P}{1 - \delta}. \quad (15)$$

Comparing the inequalities (13) and (15) shows that the latter is stricter if and only if $\Pi^{F,D} \geq \Pi^P$. This is always the case, as we demonstrate in the proof of Proposition 3. Hence,

the deviation condition (15) is not only necessary but also sufficient for formability. Note that formability decreases in T , that is, it holds for a smaller set of discount factors because the left-hand side decreases in T . This holds because an increase in T , which only affects the left-hand side of condition (15), shifts the weight Π^C to the smaller term Π^F .

Renegotiation-Proofness. Collusion is opportunism-proof if the manufacturers have no incentive to deviate jointly from the collusive price. Suppose the manufacturers jointly behave opportunistically in the present period. They can earn an opportunism profit of Π^{JD} by lowering w_i and increasing F_i for each retailer. As a result, the retailers believe in competition in the next period. Confronted with competitive beliefs, the re-formation phase starts so that the manufacturers need to play w^C for T periods to convince retailers of collusive prices again. This yields the following condition of opportunism-proofness:

$$\frac{\Pi^C}{1-\delta} \geq \Pi^{JD} + \delta \left(\frac{(1-\delta^T)}{1-\delta} \Pi^F + \frac{\delta^T}{1-\delta} \Pi^C \right). \quad (16)$$

Collusion is formable and opportunism-proof if the manufacturers are sufficiently patient, that is, if condition (15) for formability and condition (16) for opportunism-proofness hold. Increasing the adaptation length T of the beliefs makes collusion less formable but more opportunism-proof, that is, relaxes condition (16), but tightens condition (15). Opportunism-proofness is harder to satisfy than stability whenever collusion is formable. To see this, compare the right-hand side of condition (16) with the stability condition (12) and note that $\Pi^{JD} > \Pi^D$.

Punishment is not renegotiation-proof whenever collusion is formable. To see this, note that whenever the formability condition holds, and the manufacturers are supposed to punish, they are better off forming collusion again. This implies that formation dominates competitive pricing in the punishment phase and manufacturers prefer to enter the cooperative phase again.

Proposition 3. *Suppose retailers hold adaptive beliefs and manufacturers play grim-trigger strategies. Collusion is stable if condition (12) holds, which is equivalent to the stability condition when manufacturers and retailers are pairwise integrated. Collusion is formable if condition (15) holds, which ensures that manufacturers do not deviate during formation. Collusion is opportunism-proof for δ sufficiently large if and only if $\Pi^{JD} - \Pi^C < T(\Pi^C - \Pi^F)$ holds. Punishment is renegotiation-proof if and only if condition (13) fails. That is, punishment is not renegotiation-proof whenever collusion is formable, and then the equilibrium is also not strongly renegotiation-proof. An increase in the recovery rate ρ weakly decreases T and thereby facilitates formability but impedes opportunism-proofness.*

Proof. See Appendix B. □

The takeaway is that there exists an equilibrium with adaptive beliefs in which collusion is formable, stable, and opportunism-proof. Formability requires that beliefs can adapt to

collusion after a history of competition or punishment. With such adaptive beliefs, formability and renegotiation-proof punishment are in conflict, because whenever an adaptive belief features formability, punishment is not renegotiation-proof. An interesting observation is that, for adaptive beliefs, there is a trade-off between opportunism-proofness and formability: The longer beliefs take to adapt, the harder it is to start collusion, whereas opportunistic behavior that counts on restarting collusion becomes less of a problem.

However, there is a simple strategy for manufacturers that yields a strongly renegotiation-proof equilibrium that can be stable and formable. The strategy is to collude in every period without an explicit punishment phase. Intuitively, the manufacturers do not need to punish each other as it is sufficient that retailers react with their beliefs to a manufacturer deviation. Because manufacturers do not punish, the problem of renegotiation-proofness in the punishment phase vanishes.

We now turn to the special case. Consider the degenerate strategy in which manufacturers always collude and never punish. If manufacturers play “always collusion”, the incentive constraint for stability is

$$\frac{\Pi^C}{1-\delta} \geq \Pi^D + \delta \left(\frac{1-\delta^T}{1-\delta} \Pi^F + \frac{\delta^T}{1-\delta} \Pi^C \right). \quad (17)$$

Note that the stability is supported by the beliefs because deviation requires a new formation of collusion of length T . While this strategy is the least stable strategy, within the class of trigger strategies considered here, it is the only strategy that features both formability and, in addition, features renegotiation-proof punishment.¹⁵

Proposition 4. *Suppose retailers hold adaptive beliefs. Under the strategy “always collusion,” a strongly renegotiation-proof equilibrium exists if the opportunism-proofness condition (16) holds, which also implies stability. Collusion is also formable if $\delta^T (\Pi^C - \Pi^F) \geq \Pi^{F,D} - \Pi^F$ holds.*

Proof. See Appendix B. □

6 Summary and Discussion

6.1 Empirical Implications

Our analysis yields several empirical implications that distinguish the formation of upstream collusion from its stability once established. A first implication is that the determinants of cartel formation need not coincide with those of cartel duration. Standard repeated-game factors such as firms’ patience, deviation gains, monitoring possibilities, and punishment primarily determine whether an established cartel can be sustained. In our setting, these conditions are not sufficient: manufacturers must additionally induce retailers to act as if

¹⁵This corresponds to the degenerate trigger strategy with no punishment periods; in the notation of Appendix A.1, this is $\kappa = 1$.

comparable collusive terms have been offered to their rivals. Variables that reduce downstream uncertainty should therefore be particularly relevant for cartel onset and re-formation, whereas variables affecting conventional deviation incentives should be more closely related to cartel stability. Empirically, this suggests treating cartel formation and cartel survival as distinct outcomes rather than assuming that the same factors facilitate both.

Second, successful upstream collusion may be preceded by unsuccessful or only partially successful attempts to raise wholesale prices. If retailers initially expect their rivals to continue receiving competitive terms, accepting a higher wholesale price puts them at a competitive disadvantage. Manufacturers may therefore coordinate on higher wholesale prices without immediately inducing the retail-price adjustment required for the intended collusive outcome. In data containing both wholesale and retail prices, this mechanism may appear as repeated or staggered wholesale-price increases with weak, heterogeneous, or delayed downstream pass-through before a later transition to more uniform price adjustment. Such a pattern may reflect a downstream implementation problem rather than initially weak incentives to collude.

Third, our adaptive-belief specification implies an asymmetry between cartel breakdown and re-formation. When deviations are detected quickly, retailers can rapidly lose confidence that collusive terms continue to apply throughout the market. Rebuilding that confidence may take several periods of consistent conduct. Cartel breakdown can therefore be substantially faster than cartel re-formation even when manufacturers would jointly prefer to return immediately to the collusive outcome. Empirically, one may observe abrupt breakdowns followed by longer competitive spells and a gradual re-convergence of wholesale and retail prices. This suggests examining the duration of post-breakdown competitive spells and the likelihood of re-formation separately from the likelihood of cartel breakdown.

Finally, adaptive beliefs generate a distinctive comparative-static prediction. Faster adjustment of downstream beliefs facilitates cartel formation because retailers need less time to become convinced that a common change in upstream conduct has occurred. At the same time, faster adjustment increases the scope for joint opportunism because manufacturers can restore the collusive outcome more quickly after exploiting retailers' beliefs. Factors associated with faster downstream learning may therefore increase the incidence or speed of cartel formation without necessarily increasing cartel duration. More generally, the same variable may have different, and potentially opposing, effects on the formation and robustness of collusion. This provides an empirical implication that would be difficult to capture in a framework in which the same forces uniformly facilitate all dimensions of collusion.

6.2 Policy Discussion

These empirical predictions suggest a corresponding distinction for competition policy. In vertically related markets, an authority investigating possible upstream collusion should not only investigate whether manufacturers had the ability and incentive to reach an agreement, monitor compliance, and punish deviations. It should also investigate how the intended

collusive outcome was made credible to retailers. When contracts are bilateral and private, a retailer asked to accept a higher wholesale price may fear that competing retailers continue to receive more favorable terms. Practices that reduce this uncertainty can therefore facilitate the implementation of upstream collusion even if they do not directly improve monitoring among the upstream firms.

Vertical communication. Communication between manufacturers and retailers is one potential mechanism. Such communication may have legitimate commercial purposes and is not anticompetitive per se. Our analysis nevertheless highlights an additional channel through which it can matter for collusion. Information that reassures a retailer that competing retailers face comparable input-price increases can change the retailer’s response to a wholesale-price offer. The relevant information need not concern a common retail price. It may instead concern whether rival retailers have received similar terms, whether a wholesale-price increase is expected to be industry-wide, or whether competitors are expected to adjust their retail prices in response.

Vertical pricing practices. Vertical pricing practices can serve a related function. Resale price maintenance, recommended retail prices, minimum advertised prices, or other observable pricing policies may make it easier for retailers to infer that competitors will adjust their prices in response to higher input costs. Existing theories have identified several channels through which such practices can facilitate collusion. Our mechanism is complementary: an observable vertical pricing policy can reassure retailers that accepting a higher wholesale price will not leave them exposed to competitors that continue to price on the basis of lower input costs.

The distinction between private and observable practices is important here. A vertical restriction that is known only to one manufacturer--retailer pair does little to inform that retailer about the conditions faced by its rivals. By contrast, a common or publicly observable pricing policy can help create a shared understanding of how the downstream market will respond. The German coffee cartel provides a suggestive example. Wholesale-price coordination was initially relatively ineffective, while the arrangement became more successful after resale price maintenance was introduced (Holler and Rickert, 2022). Our analysis provides one interpretation of this pattern: the vertical practice may have helped solve the downstream implementation problem by making coordinated retail-price adjustment more credible.

Hub-and-spoke communication. Similar reasoning applies to hub-and-spoke arrangements. The existing literature emphasizes their role in transmitting information among horizontal competitors, coordinating conduct, and facilitating monitoring (Harrington, 2018; OECD, 2019; Garrod et al., 2021). Our analysis suggests an additional role for the vertical intermediary. A hub may reassure a downstream firm that the terms or conduct it observes are not specific to its own bilateral relationship but form part of a broader market-wide change.

Such communication can facilitate cartel formation by reducing strategic uncertainty downstream.

This distinction may be relevant even when there is no explicit downstream agreement on retail prices. Communication that allows retailers to infer that their rivals face comparable wholesale terms may be sufficient to change their pricing incentives. Competition authorities should therefore consider whether a hub’s informational role helped implement upstream coordination, rather than focusing exclusively on whether it transmitted an explicit horizontal agreement.

Public signals and common cost narratives. Public information may also affect the downstream implementation constraint. Industry announcements, trade-association statements, trade-press reports, or common narratives about cost increases can influence retailers’ expectations about the terms faced by their competitors. This may be particularly relevant during periods of substantial common cost shocks. If retailers already expect an industry-wide increase in wholesale prices, they may be less reluctant to accept a higher offer from their own supplier.

This implication should be interpreted cautiously. Common cost shocks naturally generate parallel price movements, and public discussion of such shocks may be entirely legitimate. Our model does not imply that parallel price increases or public cost information are evidence of collusion. The more specific implication is that such information can affect the implementation of a collusive increase by changing retailers’ expectations. In an investigation, the relevant question is therefore whether the information was used merely to communicate genuine cost developments or also to create confidence that supra-competitive price increases would be implemented throughout the market.

Screening formation separately from stability. More generally, our results suggest distinguishing between evidence that explains why a cartel could be *sustained* and evidence that explains how it was *formed*. Traditional plus factors such as repeated interaction, monitoring opportunities, and credible retaliation primarily bear on sustainability. In vertically related markets with private contracting, authorities may additionally look for evidence showing how downstream uncertainty was overcome.

Potentially informative patterns include unsuccessful wholesale-price increases followed by a successful coordinated increase, downstream resistance followed by reassurance about competitors’ terms, and renewed vertical communication shortly before collusion re-forms after a period of competition. None of these observations establishes collusion in isolation. Their relevance is that they are consistent with a specific economic mechanism: upstream firms may need to persuade downstream firms that higher input prices apply sufficiently broadly for the intended collusive outcome to be implemented.

The policy implication is therefore narrower than a general presumption against vertical communication or vertical restraints. These practices often serve legitimate and potentially efficiency-enhancing purposes. Our analysis instead identifies an additional question for car-

tel investigations. When upstream firms sell through independent retailers under private contracts, competition authorities should ask not only how the manufacturers coordinated and disciplined one another, but also how they induced retailers to believe that competing retailers faced comparable terms. The answer may be important for understanding both how collusion was formed and why it became effective.

7 Conclusion

Upstream collusion in vertically related markets involves more than the conventional incentive problem among cartel members. When manufacturers contract privately with competing retailers, they do not control final-market outcomes directly. They must induce retailers to accept the contracts and set the prices required to implement the intended collusive outcome. Secret contracting therefore creates a downstream implementation constraint: even manufacturers that are willing and able to coordinate may fail to implement collusion if retailers do not expect their rivals to face comparable terms. This leads to a distinction between the sustainability of an established cartel and its formability, that is, whether the collusive outcome can be reached from a non-collusive history.

Our results show that retailer beliefs are central to both dimensions. With efficient two-part tariffs and non-reactive beliefs, supra-competitive collusion cannot be sustained, irrespective of manufacturer patience. When retailers instead anticipate manufacturers' grim-trigger strategies, collusion can be sustainable and even renegotiation-proof if manufacturers are sufficiently patient and the relevant opportunism condition is satisfied. Yet the same beliefs that make an established cartel robust prevent it from forming after a competitive history. Retailers who expect continued competitive conduct will not implement the contracts required for a transition to the collusive outcome.

Adaptive beliefs can overcome this formation problem, but they generate a trade-off between formation and robustness. When retailers gradually interpret repeated collusive wholesale-price offers as evidence of an industry-wide change in conduct, manufacturers can build the confidence required to move from competition to collusion. Faster belief adjustment facilitates this transition because retailers become willing to implement the collusive contracts more quickly. At the same time, it increases the scope for joint opportunism because manufacturers can restore the collusive outcome more rapidly after exploiting retailers' beliefs. Moreover, whenever adaptive beliefs make collusion formable under a conventional punishment strategy, the prescribed competitive punishment is not renegotiation-proof: manufacturers would prefer to start rebuilding collusion before the punishment phase is completed.

A particularly simple strategy highlights a broader implication of this mechanism. Manufacturers need not always discipline deviations through an explicit punishment phase. If they return immediately to the collusive wholesale price after a deviation, the deviation can nevertheless reset retailers' beliefs and force the manufacturers to rebuild retailer confidence before the collusive outcome is again implemented. Under suitable conditions, this belief-

reset cost is sufficient to discipline both unilateral and joint deviations and can support a strongly renegotiation-proof and formable equilibrium. The same downstream implementation friction that can obstruct cartel formation can therefore become a source of cartel discipline once manufacturers attempt to maintain collusion.

These results suggest that the empirical determinants of cartel formation need not coincide with those of cartel stability. In vertically related markets with private contracting, unsuccessful attempts to increase wholesale prices, delayed retail-price responses, or persistent competitive phases after cartel breakdown may reflect difficulties in establishing the downstream beliefs required for implementation rather than an absence of upstream coordination. Conversely, practices that reduce downstream uncertainty—such as vertical communication, observable pricing practices, hub-and-spoke communication, or public signals about industry-wide conditions—may facilitate the formation or re-formation of collusion by reassuring retailers that comparable terms apply to their rivals. Such practices often have legitimate purposes and should not themselves be interpreted as evidence of collusion. Our analysis instead identifies an additional question for competition policy: alongside asking how upstream firms coordinated, monitored, and disciplined one another, authorities should ask how they induced downstream firms to implement the intended collusive outcome.

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A Appendix: Extensions

We extend the baseline model along several dimensions. First, we allow for different definitions of beliefs that fall outside our parameterization. For example, we consider manufacturers playing more general trigger strategies and define the corresponding belief system, with grim-trigger strategies and beliefs arising as a limiting case. Second, we relax the assumption of within-period passive beliefs and consider symmetric beliefs. Finally, in the third subsection we extend the analysis of non-reactive beliefs to linear wholesale pricing.

A.1 Trigger Beliefs

In this section, we define trigger beliefs and derive the conditions for the sustainability and other properties of collusion. Following a deviation in a collusive period, manufacturers play the punishment action for $\kappa - 1$ periods and return to collusion in the κ -th period after the deviation.

We define trigger strategies as in Green and Porter (1984): Define period t to be collusive if

- (a) $t = 0$, or
- (b) period $t - 1$ was collusive and $w_{it-1} = w^C$ for all i , or
- (c) period $t - \kappa$ was collusive and $w_{it-\kappa} \neq w^C$ for some i .

Define period t to be competitive otherwise. Manufacturer i sets

$$w_{it} = \begin{cases} w^C & \text{if } t \text{ is collusive,} \\ w^P & \text{if } t \text{ is competitive.} \end{cases}$$

Assume that the collusive price equals the monopoly price ($w^C = w^M$), as defined in Subsection 4.3. Any deviation by one manufacturer causes the other manufacturer to carry out a punitive action of $w^P = 0$. Again, $w^P = 0$ must hold because a deviation during the reversionary periods is not punished as the future actions and beliefs are fixed. Hence, the punishment action must be the same as the short-term optimal action, that is $w^P = 0$ as we demonstrated before. This maximizes the manufacturers' profits because it aligns the incentives of retailer and manufacturer. Similar to the previous section, trigger beliefs emerge when considering a weak PBE in which manufacturers use trigger strategies instead of grim-trigger strategies.

Definition 5. (Trigger beliefs)

Choose a collusive price level p^C and a punishment price level p^P . Formally, the retailers' beliefs correspond to the manufacturer strategies for collusive and competitive periods as defined above:

$$p_{-it}^e = \begin{cases} p^C & \text{if } t \text{ is collusive,} \\ p^P & \text{if } t \text{ is competitive.} \end{cases}$$

When both manufacturers play trigger strategies with actions w^C and w^P , the retailers' corresponding trigger-beliefs are correct on the collusive equilibrium path. They are also correct in the punishment phase.

Trigger beliefs assume that retailers understand the manufacturers' continuation strategy, including the timing of punishment and the return to collusion. With finite punishment, such beliefs require a richer public state than the binary regime in Definition 1, for example a punishment clock or a state space that distinguishes collusion from each remaining punishment period.

Qualitatively, our insights for grim-trigger strategies carry over to more general trigger strategies. Trigger strategies with limited punishment imply stricter conditions for stability and opportunism-proofness compared to grim-trigger strategies. Nevertheless, they may be attractive for reasons not modeled here, including cost and demand shocks as well as other uncertainty that could result in unwarranted punishment, which, under grim-trigger strategies, would be very costly. We summarize our findings in the following proposition:

Proposition 5. *With trigger beliefs, there exists an equilibrium in which manufacturers are able to sustain collusion on the industry-profit-maximizing wholesale price (with trigger strategies) if the discount factor is high enough to satisfy condition (18). This condition is equivalent to the incentive constraint when manufacturers and retailers are pairwise integrated. Furthermore, the condition requires more patience than under grim-trigger beliefs and strategies. Collusion is opportunism-proof only if the stricter condition (19) holds. Punishment is always renegotiation-proof, and collusion is never formable.*

Proof. Consider the conditions that are needed to sustain collusion. After any deviation by a manufacturer, the beliefs revert to $p_{-i}^e = p^P$ for $\kappa - 1$ periods, which results in profits of $\Pi^P = p^P \cdot D_i(p^P, p^P)$. After $\kappa - 1$ punishment periods, however, the retailers believe in collusion at a price of p^C again. Collusion is sustainable when no manufacturer wants to deviate from the trigger strategy given by the incentive condition:

$$\frac{\Pi^C}{1 - \delta} \geq \Pi^D + \delta \left(\frac{1 - \delta^{\kappa-1}}{1 - \delta} \Pi^P + \frac{\delta^{\kappa-1}}{1 - \delta} \Pi^C \right). \quad (18)$$

This condition is more difficult to fulfill than the incentive condition (9) for grim-trigger strategies and beliefs. The punishment in condition (18) is less harsh and lasts for $\kappa - 1$ periods, such that the expression on the right-hand side is larger than in the condition with grim-trigger strategies. Recall that manufacturers make the same profits as pairwise integrated manufacturer-retailer pairs (Proposition 2). Note that the individual profits, Π^C , Π^D , and Π^P , are still identical to an integrated firm's profits. This implies that the stability condition (18) is the same for vertically separated and vertically integrated manufacturer-retailer pairs whenever they play trigger strategies of length κ .

To check whether the equilibrium is opportunism-proof, we must consider a revised version of condition (11) that applies to trigger beliefs. Because trigger beliefs are forgiving after $\kappa - 1$

periods, a joint deviation of both manufacturers is not “punished by the beliefs” forever.¹⁶ Hence, the conditions can be written as

$$\frac{\Pi^C}{1-\delta} \geq \Pi^{JD} + \delta \left(\frac{1-\delta^{\kappa-1}}{1-\delta} \Pi^P + \frac{\delta^{\kappa-1}}{1-\delta} \Pi^C \right). \quad (19)$$

Again, the opportunism-proofness condition (19) resembles the stability condition (18), except that $\Pi^{JD} > \Pi^D$, which makes the condition harder to meet. If condition (19) holds, collusion is robust against joint deviations by the manufacturers, that is, opportunistic behavior.

The punishment is renegotiation-proof because, in a punishment phase, both retailers believe $w^P = 0$ to which the best response is $w = 0$ as well. Beliefs do not depend on the current wholesale price during punishment. Thus, the argument in Section 4.3 applies, such that focusing on short-term best responses is valid.

To see that collusion is not formable with trigger beliefs, consider an argument by contradiction. Formability requires that the PBE under consideration can arise after an arbitrary history of the game. Let us consider a specific history. Suppose that, for the first κ periods, the manufacturers do not collude. With this history, all the following periods starting with period $\kappa + 1$ are labeled as non-collusive. It follows that it is impossible for retailers to hold collusive beliefs at any future point of the game. Recall that periods are only labeled collusive after the first period if in period t , the previous period $t - 1$ was collusive or period $t - \kappa$ was collusive. $\square$

The derived equilibrium with trigger beliefs is Pareto-efficient and features renegotiation-proof punishment. If, in addition, the manufacturers are patient enough for the condition of opportunism-proofness to hold, the equilibrium is strongly renegotiation-proof.

Corollary 2. *The equilibrium with trigger beliefs described in Proposition 5 is strongly renegotiation-proof if the opportunism-proofness condition (19) is satisfied.*

A.2 Symmetric Beliefs

After considering passive beliefs, we turn to the analysis of the case that retailers have symmetric beliefs. Whereas passive beliefs do not condition on the retailer’s own current offer, a retailer with symmetric beliefs infers from its own offer that the rival retailer has received the same one. Such beliefs therefore violate within-period passivity and depend on the current private offer rather than on the public history alone. For this section we relax that requirement and again focus on weak PBE. We analyze these beliefs because they are canonical in the secret-contracting literature (Pagnozzi and Piccolo, 2012; Liu and Thomes, 2021) and because they represent the polar case in which retailers immediately interpret a wholesale-price change as industry-wide. Symmetric beliefs are defined as follows:

¹⁶By contrast, the grim-trigger beliefs switch forever to the competitive price level in response to a deviation. This effectively punishes the manufacturers that face pessimistic retailers from then on.

Definition 6. (Symmetric Beliefs) The price expectation p_{-it}^e of retailer i in period t about the price of retailer $-i$ is $p_i(w_{it}, p(w_{it}))$.

In other words, when a retailer receives an unexpected offer deviating from the candidate equilibrium, the retailer revises its belief and believes that its rival has received the same offer by its manufacturer. Symmetric beliefs are independent of past histories because they only rely on the information contained in the current wholesale price offer. We assume that manufacturers play (grim-)trigger strategies as in the previous sections. The case of symmetric beliefs is also analyzed in Liu and Thomes (2021) who consider the linear demand case and, hence, offer closed-form solutions for the critical discount factor. Symmetric beliefs may be particularly plausible if retailers expect that manufacturers coordinate their actions and expect that they do this in a symmetric way.

We focus again on industry-profit-maximizing collusion that naturally arises when manufacturers jointly maximize their profits given symmetric beliefs. Denote the price expectation of retailer i with symmetric beliefs by p_{it}^e as above. This allows the manufacturer to essentially choose the symmetric price level, such that the joint maximization problem of the manufacturers can be rewritten as

$$\Pi^C := \frac{1}{2} \max_p p \cdot D_i(p, p) + p \cdot D_{-i}(p, p) = \frac{\Pi^M}{2}.$$

Hence, the jointly optimal wholesale price induces the industry-profit-maximizing retail price. Moreover, this implies that any joint deviation by the manufacturers will always be the industry-profit-maximizing price if retailers hold symmetric beliefs. In contrast to (grim-)trigger beliefs, where the optimal joint deviation of the manufacturers is to charge a wholesale price below zero, such a deviation is not optimal because only the collusive price maximizes the manufacturers' profits. Because the manufacturers collude at the Pareto-efficient level, collusion is opportunism-proof. Additionally, punishment is not renegotiation-proof because the manufacturers would prefer to revert to collusion in every punishment period.

Collusion is formable because symmetric beliefs instantly adapt to the new wholesale price in every period. Manufacturers only need to agree on the collusive price and set it in any period. In a period in which the manufacturers set wholesale prices of w^C , the retailers' expectations are immediately equal to $p_i(w^C, p(w^C)) = p_i(w^C, p^C) = p^C$, which corresponds to the expectation of collusion. Forming collusion immediately leads to stable collusion as long as the stability condition is satisfied.

From Pagnozzi and Piccolo (2012), we know that symmetric beliefs affect competition between vertically separated manufacturers. Competition is less fierce due to a so-called belief effect, which increases the competitive wholesale price above marginal costs. If punishment, however, relies on the competitive wholesale prices and profits, the stability of collusion is affected. Manufacturers must be more patient to satisfy the condition of stable collusion.

Symmetric beliefs violate the “no-signaling-what-you-don’t-know” Condition 1.¹⁷ We thus

¹⁷Pagnozzi and Piccolo (2012) discuss this observation and also show that, with common cost shocks,

look for a perfect Bayesian equilibrium in the context of symmetric beliefs:

Proposition 6. *With symmetric beliefs, there exists a collusive equilibrium with (grim-)trigger strategies if condition (21) holds. Collusion is also formable if the stability condition holds. Collusion is always opportunism-proof, but punishment is not renegotiation-proof. If condition (21) holds, both the competitive equilibrium and the collusive equilibrium are not renegotiation-proof.*

Proof. Let us first consider the stability condition. As argued in the text before the proposition, manufacturers collude on the industry-profit-maximizing level and earn profits of $\Pi^C = \Pi^M/2$. If the manufacturers make a one-sided deviation from the collusive agreement, they must consider that changing their wholesale price also influences the belief of their own retailer. A deviating manufacturer maximizes the following problem:

$$w^D = \arg \max_{w_{it}} w_{it} \cdot D_i(p_i(w_{it}, p(w_{it})), p_{-i}(w^C, p(w^C))) \\ + (p_i(w_{it}, p(w_{it})) - w_{it}) \cdot D_i(p_i(w_{it}, p(w_{it})), p(w_{it})).$$

This results in a profit Π^D for the deviating manufacturer. Note that $w^D > 0$. Punishment is assumed to be carried out on the competitive wholesale price level. The competitive benchmark corresponds to the case analyzed in Pagnozzi and Piccolo (2012). Under symmetric beliefs, manufacturers solve

$$\max_{w_i} w_{it} \cdot D_i(p_i(w_{it}, p(w_{it})), p_{-i}(w_{-it}, p(w_{-it}))) + (p_i(w_{it}, p(w_{it})) - w_{it}) \cdot D_i(p_i(w_{it}, p(w_{it})), p(w_{it})).$$

This results in Π^P . By using the envelope theorem, the first-order condition simplifies to

$$\underbrace{w_i \frac{\partial D_i(\cdot)}{\partial p_i}}_{<0} + \underbrace{(p(w_i, p(w_i)) - w_i) \cdot \frac{\partial D_i(\cdot)}{\partial p_{-i}}}_{\text{belief effect: } >0} = 0. \quad (20)$$

Applying symmetry to equation (20) defines the equilibrium wholesale prices w^P . Note that the second term of (20) is positive at $w_i = 0$, which implies that $w^P > 0$. Under competition, with symmetric beliefs, prices are above the price level under competition with passive beliefs. The stability condition is given by

$$\frac{\Pi^C}{1 - \delta} \geq \Pi^D + \frac{\delta}{1 - \delta} \Pi^P. \quad (21)$$

To see why collusion is formable with symmetric beliefs, recall the definition of formability. Symmetric beliefs allow for forming collusion if there exists a strategy profile, such that for this belief, best responses are played in period $t \geq s$, and there exists a weak PBE that results in payoff V^C . Given any history before period s – the period in which collusion is about to be

symmetric beliefs can be consistent with a PBE.

formed –, manufacturers play mutually best responses in the following periods when setting wholesale prices that result in the collusive price defined by Equation (3). The joint profit maximization of both manufacturers and setting w^C is a weak PBE if the stability condition for collusion – Equation (21) – holds. In this equilibrium, both manufacturers earn a payoff of $V^C = \Pi^C/(1 - \delta)$. Hence, collusion is formable according to our definition.

As in Section 5.3, we ask whether there exist incentives to deviate from formation. Due to symmetric beliefs, collusion is in place directly after it is formed, such that the condition for deviating from formation is identical to deviating from collusion in equation (21).

To check whether collusion is opportunism-proof, we consider the joint maximization problem of the manufacturers. As shown in the text, this leads to the Pareto-efficient wholesale price level. That is, manufacturers always prefer to set w^C when jointly maximizing. Thus, there is no scope for opportunistic behavior because the beliefs directly react to any change in wholesale prices. Following the same argument, punishment is not renegotiation-proof because manufacturers would prefer to renegotiate and jointly revert to setting w^C .

The competitive equilibrium is not renegotiation-proof whenever a collusive equilibrium with higher manufacturer profits exists. Hence, if Condition 21 holds, there exists a Pareto-dominant equilibrium for manufacturers that can be reached by a joint deviation of the manufacturers. With symmetric beliefs, retailers are immediately willing to accept the collusive contract expecting wholesale prices to increase in the whole industry. $\square$

With symmetric beliefs, there is no opportunism problem because lowering the wholesale prices negatively affects the belief such that retailers do not accept collusive contracts. Because the symmetric belief follows the manufacturers' actions, there is no long-term response of retailers and thus retailers do not discipline the manufacturers as, for example, the retailers' response does when they are holding trigger beliefs. This implies that formation, in the sense of needing to convince retailers, is not an issue with symmetric beliefs. As a downside for collusion, symmetric beliefs do not support the renegotiation-proofness of punishment because they make the punishment phase prone to renegotiation incentives. The reason is that the retailers always accept the collusive contracts, believing that wholesale prices are at the collusive level in the whole industry. During a punishment phase both manufacturers are able to renegotiate a joint increase of their wholesale prices again, which in turn will be accepted by each retailer. Therefore, manufacturers are able to return to the Pareto-dominant collusive equilibrium. This intuition also applies to the competitive equilibrium given that the manufacturers' discount factor is sufficiently large to potentially support collusion.

A.3 Linear Wholesale Prices

We extend the case of non-reactive beliefs to linear tariffs. For this extension we restrict the manufacturers' contracts to be linear. Each manufacturer offers a contract that only includes a wholesale price w_i .

Let us reconsider the case of non-reactive beliefs. For two-part tariffs we have shown that there is no collusive equilibrium. In contrast, for linear tariffs we construct grim-trigger

equilibria and show that for sufficient patience an equilibrium with collusive prices above the competitive price level always exists. Let us first define the competitive equilibrium. The symmetric one-shot equilibrium with correct retailer beliefs is characterized by $p^N = p_i(w^N, p^N)$ and

$$w^N = \arg \max_w w D_i(p_i(w, p^N), p^N), \quad i \in A, B.$$

The resulting manufacturer profit is $\Pi^N = w^N D_i(p^N, p^N)$. Playing w^N in each period and retailers believing p^N is a weak PBE of the repeated game.

Consider that manufacturers collude symmetrically at a wholesale price of w^C using grim-trigger strategies. We denote the wholesale prices in punishment periods by w^P . A deviating manufacturer chooses a wholesale price denoted by w^D .

If there exists an equilibrium with a collusive price level of w^C , the beliefs have to be correct on the equilibrium path and equal $p^e = p^*(w^C, p^e)$. We construct such an equilibrium with time-constant beliefs to show the existence of a collusive equilibrium. If manufacturer A deviates in a period, the belief of the retailer of the non-deviating manufacturer B remain incorrectly fixed at p^e . Similarly, in punishment periods, both retailers may hold incorrect beliefs, expecting the price to be at the collusive level, whereas the actual price level depends on w^P that is generally not equal to w^C . Unlike the case of two-part tariffs, incorrect retailer beliefs do not render collusion impossible with linear tariffs as the retailers mechanically accept different wholesale price levels, so that the effect of the wholesale prices on manufacturer profits is preserved.

Let us define the profits necessary to evaluate the stability of collusion:

$$\Pi^C \equiv w^C D_i(p^e, p^e),$$

$$\Pi^D = \max_w w \cdot D_i(p^*(w, p^e), p^e).$$

Mutual best responses in the punishment phase imply

$$\begin{aligned} w^P &= \arg \max_w w \cdot D_i(p^*(w, p^e), p^*(w^P, p^e)), \\ \Pi^P &= w^P \cdot D_i(p^*(w^P, p^e), p^*(w^P, p^e)), \end{aligned}$$

where $p^*(w^P, p^e) < p^e$ holds for any $w^P < w^C$.

Assumption 2. *We assume that at the locus at $w^C = w^D = w^P = w^N$, the implicitly defined wholesale prices w^N, w^D , and w^P are unique and $0 < \frac{\partial w^D(w^C)}{\partial w^C} < 1$ and $0 < \frac{\partial w^P(w^C)}{\partial w^C} < 1$ hold.*

This assumption holds globally for linear demand. The assumption implies that the wholesale prices have the usual ordering in a neighborhood around the competitive price level: If $w^C > w^N$, that is, if there is effective collusion, then $w^C > w^D > w^P > w^N$ holds (see proof of Proposition 7).

An equilibrium with grim-trigger strategies exists if there is a critical discount factor below one, such that the stability condition holds. This can be rewritten as

$$\Pi^D - \Pi^C < \Pi^C - \Pi^P. \quad (22)$$

Proposition 7. *Suppose that retailers hold non-reactive beliefs and Assumption 2 holds. With linear wholesale prices, there is a collusive equilibrium with $w^C > w^N$ if manufacturers are sufficiently patient. This collusive equilibrium is also formable.*

Proof. If $0 < \frac{\partial w^D(w^C)}{\partial w^C}|_{w^C=w^D} < B1$ holds, then increasing w^C implies $w^C > w^D$. Then, $w^C > w^N$ (by construction) implies that $w^D > w^N$. Using $w^C > w^D$, we show that

$w^D > w^P$ holds: Note that $w^D > w^P \Leftrightarrow p^e > p^*(w^P, p^e) \Leftrightarrow w^P < w^C$. The last inequality follows again from the argument that starting at $w^C = w^N$, increasing w^C implies by $\frac{\partial w^P(w^C)}{\partial w^C} < 1$ that $w^C > w^N$ implies $w^C > w^P$. Hence, $w^C > w^P$ implies that $w^D > w^P$ must hold. Rewrite condition (22) as

$$2\Pi^C - \Pi^P - \Pi^D > 0.$$

This can be rewritten as

$$\begin{aligned} & \int_{w^P}^{w^C} wD(p^*(w, p^e), p^*(w, p^e))dw + \int_{w^D}^{w^C} wD(p^*(w, p^e), p^e)dw \\ &= \int_{w^P}^{w^D} wD(p^*(w, p^e), p^*(w, p^e))dw + \int_{w^D}^{w^C} w[D(p^*(w, p^e), p^e) + D(p^*(w, p^e), p^e)]dw. \end{aligned} \quad (23)$$

Note $2\Pi^C - \Pi^P - \Pi^D = 0$ holds at $w^C = w^N$. Thus if, at this locus, the derivative of (23) with respect to w^C is positive, then the stability condition holds for some $w^C > w^N$ and some $\delta < 1$. Thus, there is a collusive equilibrium if, at $w^C = w^N$,

$$\partial \left(\int_{w^P}^{w^D} wD(p^*(w, p^e), p^*(w, p^e))dw + \int_{w^D}^{w^C} w[D(p^*(w, p^e), p^e) + D(p^*(w, p^e), p^e)]dw \right) / \partial w^C > 0.$$

Note that, at $w^C = w^N$, it holds that $w^D = w^P = w^N = w^C$. By Leibniz's rule we have

$$\begin{aligned} & \partial \left(\int_{w^P}^{w^D} wD(p^*(w, p^e), p^*(w, p^e))dw + \int_{w^D}^{w^C} w[D(p^*(w, p^e), p^e) + D(p^*(w, p^e), p^e)]dw \right) / \partial w^C \\ &= w^D D(p^*(w^D, p^e), p^*(w^D, p^e))w^{D'} - w^P D(p^*(w^P, p^e), p^*(w^P, p^e))w^{P'} \\ &+ \int_{w^P}^{w^D} [wD(p^*(w, p^e), p^*(w, p^e))]dw + w^C [D(p^*(w^C, p^e), p^e) + D(p^*(w^C, p^e), p^e)] \\ &- w^D [D(p^*(w^D, p^e), p^e) + D(p^*(w^D, p^e), p^e)]w^{D'} + \int_{w^D}^{w^C} [w[D(p^*(w, p^e), p^e) + D(p^*(w, p^e), p^e)]]dw \end{aligned}$$

Evaluating the derivative at $w^D = w^P = w^N = w^C = w^*$ yields

$$\begin{aligned}
& w^D D(p^*(w^D, p^e), p^*(w^D, p^e)) w^{D'} - w^P D(p^*(w^P, p^e), p^*(w^P, p^e)) w^{P'} \\
& + \underbrace{\int_{w^P}^{w^D} [w D(p^*(w, p^e), p^*(w, p^e))]'}_{=0} dw + w^C [D(p^*(w^C, p^e), p^e) + D(p^*(w^C, p^e), p^e)] \\
& - w^D [D(p^*(w^D, p^e), p^e) + D(p^*(w^D, p^e), p^e)] w^{D'} + \underbrace{\int_{w^D}^{w^C} [w [D(p^*(w, p^e), p^e) + D(p^*(w, p^e), p^e)]]'}_{=0} dw.
\end{aligned}$$

Define $D^* = D(p^*(w^*, p^e), p^*(w^*, p^e)) = D(p^*(w^*, p^e), p^e)$ with $p^*(w^*, p^e) = p^e$. The above expression can then be rewritten as

$$\begin{aligned}
& w^* D(p^*(w^*, p^e), p^*(w^*, p^e)) (w^{D'} - w^{P'}) + \\
& w^* [D(p^*(w^*, p^e), p^e) + D(p^*(w^*, p^e), p^e)] (1 - w^{D'}) \\
& = w^* D^* (2 - w^{D'} - w^{P'}).
\end{aligned}$$

As $w^* D^* > 0$, this is larger than zero if and only if

$$w^{D'} + w^{P'} < 2,$$

where $w^{D'} = \frac{\partial w^D(w^C)}{\partial w^C} \big|_{w^D=w^C}$. This holds if $w^{D'} < 1$ and $w^{P'} < 1$, which holds by Assumption 2.

Formability requires that the weak PBE can be reached from an arbitrary history through a sequentially rational transition. For the collusive weak PBE, the system of beliefs in this case is simply that, starting in period t_0 , the beliefs are constant in w^C for both retailers. Thus, if the stability condition for collusion holds, collusion is a weak PBE of the continuation game starting in period t_0 (there is no transition required). $\square$

While collusive equilibria with linear tariffs and prices just above the competitive price level exist if manufacturers are sufficiently patient, linear tariffs reduce the ability of manufacturers to extract retail rents. Thus, overall, linear tariffs do not necessarily result in higher profits for manufacturers than two-part tariffs.

B Appendix: Proofs

Proof of Lemma 1. Parts (i) and (ii) (Unique mapping from wholesale to retail prices). The text after the lemma proves that retailer i 's price is uniquely determined by $p_i(w_{it}, p_{-it}^e)$. The fixed fee enters the retailer's profit additively and therefore does not affect its price setting.

Furthermore, as we employ beliefs about p_{it}^e instead of beliefs about w_{it}^e , we establish that this is equivalent in our setting if retailers have common knowledge about the belief system and expect sequentially rational choices of their rivals. More precisely, for any history $\mathcal{H}_t$ and belief w_{it}^e , there exists a unique implied belief. Expecting a sequentially rational choice implies that the belief of retailer A , p_{Bt}^e , is the result of $p_{Bt}^e = \arg \max_{p_{Bt}} \pi_B(w_{Bt}^e, w_{At}^e = w_{At}^{2e})$, where w_{At}^{2e} is the higher-order belief of retailer A about retailer B 's belief. The identity $w_{At}^e = w_{At}^{2e}$ follows from common knowledge about the belief system and Condition 1. Due to Condition 1, beliefs are within-period passive, such that the higher-order belief w_{At}^{2e} is independent of retailer A 's private information w_{At} . Common knowledge about the belief system then implies that retailers hold correct conjectures about higher-order beliefs.

At the initial history $\mathcal{H}_0$, define (w_{A0}^e, w_{B0}^e) to be the prior belief of retailer $-i$ about the wholesale price of retailer i . The following properties ensure that the beliefs are well defined:

Retailers are symmetric and have a common understanding of the game and of the belief system, including common priors.

Beliefs are within-period passive.

Given w_{-it}^e , retailer i correctly predicts w_{it}^{2e} and thus the unique $p_{-it}^e = \arg \max_{p_{-it}} \pi_{-i}(w_{-it}^e, w_{it}^{2e})$. This follows from Assumption 1, which implies that there is a unique solution to the maximization problem.

Part (iii) (Sufficiency of the wholesale-price history). Denote by $\mathcal{G}_t$ the full public history up to period $t - 1$, containing all contract offers $(w_{i\tau}, F_{i\tau})$, acceptance decisions, and retail prices for $\tau < t$, and recall that $\mathcal{H}_t$ contains the wholesale prices only. We show that it is without loss to condition beliefs on $\mathcal{H}_t$ only instead of on the full public history $\mathcal{G}_t$. We do so by induction.

(a) *Hypothesis.* Let retailers' beliefs be an arbitrary function of the full public history, $p_{-it}^e = \psi(\mathcal{G}_t)$. We show that $\mathcal{G}_t$ is a function of $\mathcal{H}_t$, such that ψ can be written as $p_{-it}^e = g(\mathcal{H}_t)$ and the restriction to $\mathcal{H}_t$ excludes no belief system.

(b) *Induction anchor.* At $t = 0$ both histories are empty.

(c) *Induction step.* Suppose $\mathcal{G}_t$ is a function of $\mathcal{H}_t$, such that $p_{-it}^e = g(\mathcal{H}_t)$, and consider the three components of period- t play.

A fixed fee that leaves the retailer a rent is strictly dominated: by Condition 1, raising F_{it} does not affect p_{-it}^e , such that the manufacturer gains. Restricting attention to undominated offers, the participation constraint binds and F_{it} is a function of $(w_{it}, \mathcal{H}_t)$. At the binding fee the retailer accepts, such that acceptance decisions are constant across histories. Retail prices are $p_i(w_{it}, p_{-it}^e)$, unique by Part (i) and hence also a function of $(w_{it}, \mathcal{H}_t)$.

(d) *Conclusion.* All period- t actions are therefore functions of $(w_{At}, w_{Bt}, \mathcal{H}_t) = \mathcal{H}_{t+1}$, such that $\mathcal{G}_{t+1}$ is a function of $\mathcal{H}_{t+1}$. The argument applies on and off the equilibrium path.

Proof of Lemma 2. In the one-shot equilibrium, the retailer's belief about the rival's retail price coincides with the manufacturer's expectation about that price, such that $p_{-i}(w_{-it}, p_{it}^e) =$

p_{-it}^e . Plugging this into the manufacturer's problem subject to (2) and using that the retailer's participation constraint binds, such that $F_{it} = (p_i(w_{it}, p_{-it}^e) - w_{it}) D_i(p_i(w_{it}, p_{-it}^e), p_{-it}^e)$, and simplifying yields

$$\Pi_{it}(w_{it}, p_{-it}^e) = p_i(w_{it}, p_{-it}^e) \cdot D_i(p_i(w_{it}, p_{-it}^e), p_{-it}^e).$$

Maximizing with respect to w_{it} gives

$$\begin{aligned} \frac{\partial p_i(w_{it}, p_{-it}^e)}{\partial w_{it}} D_i(p_i(w_{it}, p_{-it}^e), p_{-it}^e) + \frac{\partial D_i(p_i(w_{it}, p_{-it}^e), p_{-it}^e)}{\partial p_i} \frac{\partial p_i(w_{it}, p_{-it}^e)}{\partial w_{it}} p_i(w_{it}, p_{-it}^e) &= 0 \\ \iff \frac{\partial p_i(w_{it}, p_{-it}^e)}{\partial w_{it}} \frac{\partial D_i(p_i(w_{it}, p_{-it}^e), p_{-it}^e)}{\partial p_i} w_{it} &= 0. \end{aligned} \quad (24)$$

Because the first term on the left-hand side in the previous line is assumed to be strictly positive and the second term is assumed to be strictly negative (Assumption 1), the only solution to the first-order condition is $w_{it} = 0$. Hence $w^P = 0$, and by Assumption 1 the downstream equilibrium at $w_A = w_B = 0$ is unique, which defines the competitive retail price $p^P = p_i(0, p^P)$.

Proof of Proposition 1. Step 1 (Construction of the equilibrium). We show that it is uniquely optimal for each manufacturer to set $(w_i = 0, F_{it} = \pi_i^*(w_i = 0, p_{-it}^e))$ in each period, independent of the strategy of the other manufacturer.

The participation constraint of the retailer binds in equilibrium because otherwise the manufacturer could increase profits by raising the fixed fee without affecting p_{-it}^e , by the assumption of passive beliefs. Hence $F_{it} = \pi_i^*(w_i = 0, p_{-it}^e)$ holds.

On the equilibrium path, $p_{-it}^e = p_{-it}^*$, that is, retailers' beliefs are correct and thus identical to the manufacturer's conjectures, such that the in-period manufacturer profits can be simplified as in the stage game to

$$\Pi_{it}(w_{it}, p_{-it}^e) = p_i(w_{it}, p_{-it}^e) \cdot D_i(p_i(w_{it}, p_{-it}^e), p_{-it}^e).$$

This in-period profit is insulated from the actions of the other manufacturer, and w_{it} only affects the manufacturer's profit through the price setting of the retailer. Because the manufacturers' profit in each period, on any equilibrium path, only depends on w_{it} and the belief p_{-it}^e , the discounted equilibrium profits of a manufacturer do not depend on the strategy of the other manufacturer either.

Fixing p_{-it}^e , equation (24) implies that $w_i = 0$ is optimal, independent of w_{-it} , in each period. Hence there is an equilibrium path with each manufacturer setting $w_i = 0$ in every stage game and matching time-constant beliefs by retailers, where the time-constant retail price follows because the retail price equilibrium is unique for $w_i = w_{-i} = 0$.

Step 2 (Uniqueness). Next, we exclude other equilibrium paths in which $w_{it} \neq 0$ by contradiction. Suppose that there is an equilibrium path with $w_{it} \neq 0$ in some periods. It follows from equation (24) that each manufacturer can increase current-period profits by setting $w_{it} = 0$ and $F_{it} = \pi(0, p_{-it}^e)$, resulting in $\Pi_{it}(0, p_{-it}^e)$. In particular, in every period, manufacturer i can ensure at least a profit of

$$\Pi(0, p^e) = \max_p \pi(0, p^e) = \max_p p D_i(p, p^e),$$

which is independent of the actual wholesale price of the other manufacturer. This profit is strictly larger than the profit in any candidate equilibrium with $w' \neq 0$. Formally,

$$\Pi(w', p^e) = w' D_i(p^e, p^e) + (p^e - w') D_i(p^e, p^e) = p^e D_i(p^e, p^e) < \max_p p D_i(p, p^e) = \Pi(0, p^e),$$

where the strict inequality follows from the uniqueness of the maximizer and the strict monotonicity of p_i in w_i .

As each manufacturer can ensure this deviation profit of $\Pi(0, p^e)$ in any period independently of the other manufacturer's actions, a profitable deviation from any candidate equilibrium with $w' \neq 0$ always exists, and this deviation is immune to punishment by the other manufacturer.

Step 3 (No collusion above the competitive level). By Steps 1 and 2, every symmetric weak PBE has $w_{it} = 0$ in every period, and by Step 1 the fixed fee extracts the retailer's entire profit. The resulting retail price is p^P by Lemma 2. Collusion therefore cannot raise prices above the competitive level.

Proof of Proposition 2. Step 1 (Stability). We established in the main text that manufacturers are able to sustain collusion if equation (9) holds opportunistically.

Step 2 (Equivalence to vertical integration). To show that the incentive constraint is the same if manufacturers and retailers are pairwise integrated, it is sufficient to prove that the profits are the same, $\Pi_V^j = \Pi_I^j$ for $j = C, D, P$ (V : vertically separated, I : integrated).

(a) *Collusive profits.* Because we define Π_V^C to be half of the integrated industry-maximizing profit (4), the profit from collusion in the integrated case Π_I^C is the same.

(b) *Deviation profits.* The deviation profit Π_V^D is given by the first line of equation (5) and simplifies further in the second line because the manufacturers' true actions and retailers' beliefs are aligned. The second line, however, is equal to the maximization problem of an integrated manufacturer that maximizes profit with respect to the retail price.

(c) *Punishment profits.* The punishment profits are aligned as well, following the same argument as before. They are given by equation (7). Manufacturers set $w^P = 0$, which results in the same retail prices and profits as in a vertically integrated industry.

Step 3 (Opportunism-proofness). Derived in the main text: the optimal joint deviation is characterized by equation (10), which yields $\Pi^{JD} > \Pi^D$, such that the opportunism-proofness condition (11) is stricter than the stability condition of Step 1.

Step 4 (Renegotiation-proof punishment). With grim-trigger beliefs, any deviation leads to the belief p^P forever, such that the current actions of the manufacturers have no effect on the belief of retailers. To establish that punishment is renegotiation-proof, we check for a joint action by manufacturers that would yield a Pareto improvement for manufacturers. In each period, the best response for each manufacturer is to set $w = 0$ individually. Next, we consider the profit maximization for manufacturers when they optimize jointly during a punishment period:

$$\begin{aligned} \frac{1}{2} \max_{w_{it}, w_{-it}} & w_{it} \cdot D_i(p_i(w_{it}, p^P), p_{-i}(w_{-it}, p^P)) + [p_i(w_{it}, p^P) - w_{it}] \cdot D_i(p_i(w_{it}, p^P), p^P) \\ & + w_{-it} \cdot D_{-i}(p_{-i}(w_{-it}, p^P), p_i(w_{it}, p^P)) + [p_{-i}(w_{-it}, p^P) - w_{-it}] \cdot D_{-i}(p_{-i}(w_{-it}, p^P), p^P). \end{aligned}$$

We again assume that this profit is quasi-concave, such that we can use first-order conditions. Using the retailers' first-order conditions and applying symmetry, using that manufacturer profits are well behaved by assumption such that the optimum is symmetric, the first-order conditions can be rewritten as

$$\left[D_i(p_i(w, p^P), p_{-i}(w, p^P)) - D_i(p_i(w, p^P), p^P) \right] + w \underbrace{\frac{\partial p_i}{\partial w_{it}}}_{>0} \left[\underbrace{\frac{\partial D_i}{\partial p_i} + \frac{\partial D_{-i}}{\partial p_i}}_{<0} \right] = 0 \quad \forall i. \quad (25)$$

At $w = 0$ both terms vanish: the first because $p_{-i}(0, p^P) = p^P$, and the second because it carries the factor w . Hence $w = 0$ satisfies (25), and by quasi-concavity it is the joint optimum. Playing $w = 0$ is therefore the best manufacturers can do, such that there is no continuation game that can be reached by manufacturers and that yields larger profits.

Proof of Proposition 3. Recall from Subsection 4.3 that $\Pi^D > \Pi^C > \Pi^P$. Two further rankings are used below.

(a) $\Pi^C > \Pi^F$. In a formation period both retailers charge p^F , such that realized industry profit per pair is $p^F D_i(p^F, p^F) < \Pi^C$ by equation (3). The manufacturer receives less, because the fixed fee is set against the belief p^P : $\Pi^F < p^F D_i(p^F, p^F)$, using $p^F > w^C$ and $D_i(p^F, p^F) > D_i(p^F, p^P)$.

(b) $\Pi^{F,D} > \Pi^P$. A manufacturer deviating in a formation period faces a retailer that still expects p^P , whereas the rival retailer charges $p^F > p^P$. At $w_i = 0$ the variable part of (14) vanishes and the expression equals Π^P . By the retailer's first-order condition (1), its derivative at $w_i = 0$ equals $D_i(p^P, p^F) - D_i(p^P, p^P) > 0$ by Assumption 1. Hence $w^{F,D} > 0$

and $\Pi^{F,D} > \Pi^P$. Because w^C is not the spot best response against p^P , $\Pi^{F,D} > \Pi^F$ holds as well.

Step 1 (Stability). On the collusive path $\lambda_0 = 1$. After a deviation $\tau = 1$ sets $\lambda = 0$, and manufacturers never return to w^C , such that beliefs remain p^P forever; the relevant condition is (12). As in Subsection 4.3, $w^P = 0$ and Π^C , Π^D , Π^P are unchanged, such that equivalence to pairwise integration follows from the proof of Proposition 2.

Step 2 (Formability). From the worst history $\lambda_s = 0$, beliefs remain p^P for T periods by Remark 3, after which Step 1 applies. Definition 4 therefore requires (13) and (15). Writing both inequalities with the formation value on the left-hand side, both have the same left-hand side, and the right-hand side of the latter exceeds that of the former by $\Pi^{F,D} - \Pi^P > 0$ by (b), such that (15) is the relevant condition. Multiplying it by $(1 - \delta)$ yields

$$\delta^T (\Pi^C - \Pi^F) \geq (1 - \delta) (\Pi^{F,D} - \Pi^F) + \delta (\Pi^P - \Pi^F), \quad (26)$$

which at $\delta = 1$ reduces to $\Pi^C \geq \Pi^P$, which holds strictly because $\Pi^C > \Pi^P$. By continuity, collusion is formable for δ sufficiently large.

Step 3 (Opportunism-proofness). Multiplying (16) by $(1 - \delta)$, opportunism-proofness is equivalent to $g(\delta) \geq 0$, where

$$g(\delta) \equiv \Pi^C - (1 - \delta) \Pi^{JD} - \delta (1 - \delta^T) \Pi^F - \delta^{T+1} \Pi^C.$$

Because $g(1) = 0$, the comparison is decided by $g'(1) = \Pi^{JD} + T\Pi^F - (T + 1)\Pi^C$, such that $g(\delta) > 0$ for δ close to one if and only if

$$\Pi^{JD} - \Pi^C < T (\Pi^C - \Pi^F), \quad (27)$$

where $\Pi^C - \Pi^F > 0$ by (a). Under the reverse strict inequality, opportunism-proofness fails for all sufficiently patient manufacturers.

Step 4 (Punishment). In a punishment period, manufacturers obtain $\Pi^P / (1 - \delta)$ by continuing to punish and the right-hand side of (13) by jointly restarting formation. Punishment is thus renegotiation-proof if and only if (13) fails. Whenever collusion is formable, (15) and hence (13) hold by Step 2, such that punishment is not renegotiation-proof.

Step 5 (Comparative statics in ρ). By Remark 3, T is weakly decreasing in ρ . A lower T shifts weight on the left-hand side of (15) from Π^F to the larger Π^C and thus relaxes it, while tightening (27).

Proof of Proposition 4.

Under “always collusion” there is no punishment phase, such that weak renegotiation-proofness reduces to opportunism-proofness. A joint deviation yields Π^{JD} and leads to belief

p^P , such that the continuation consists of T re-formation periods followed by collusion: the relevant condition is (16), unchanged from the grim-trigger case. It implies stability, because the stability condition under “always collusion” differs from it only in that Π^{JD} is replaced by Π^D , and $\Pi^{JD} > \Pi^D$.

The incentive constraint for formation can be written as

$$\delta^T (\Pi^C - \Pi^F) \geq \Pi^P - \Pi^F.$$

A deviation from the collusive price in the formation phase results in a profit of $\Pi^{F,D}$. It triggers no punishment but leads to belief p^P , such that no manufacturer deviates during formation if

$$\frac{1 - \delta^T}{1 - \delta} \Pi^F + \frac{\delta^T}{1 - \delta} \Pi^C \geq \Pi^{F,D} + \delta \left(\frac{1 - \delta^T}{1 - \delta} \Pi^F + \frac{\delta^T}{1 - \delta} \Pi^C \right),$$

which simplifies to

$$\delta^T (\Pi^C - \Pi^F) \geq \Pi^{F,D} - \Pi^F. \quad (28)$$

Observe that a deviation in the formation phase is not profitable under a stricter condition than the incentive condition for formation because $\Pi^{F,D} > \Pi^P$. Additionally, a necessary condition is $\Pi^C > \Pi^{F,D}$. Because $\delta < 1$, the left-hand side of condition (28) decreases in T , such that formability becomes harder to satisfy the larger T .